# Quadratic Trade Rigidity and Cubic Orientation in Conic Matching Quotients

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**Abstract**

A perfect matching of marked points on a conic determines a product of secant lines; restricting that product to the conic forgets the pairing. Among full $\mathrm{PGL}_2(q)$-orbits of such matchings over odd finite fields, we classify those whose conic-quotient evaluation space has a one-dimensional strength-two trade—a signed relation annihilating all quadratic coordinate products—generated by a two-valued vector: exactly two such orbits occur, the balanced $B_3/\mathbb{F}_7$ and $H_3/\mathbb{F}_{11}$ orbits. Targeted modular detectors, made exhaustive by Faber's tame subgroup theorem, exclude every other orbit without a field census. From the trade we reconstruct the two complementary sheets up to interchange, assuming neither self-association nor Gorensteinness, and we orient them by the first nonzero signed moment: the signed moments vanish through degree two, and that first survivor is an anti-invariant cubic.

We prove the matching hypothesis sharp. For either surviving stabilizer the fixed locus in the ambient conic-product fiber is an affine line of $q$ pairwise nonconjugate rational points: $q-2$ of them are nonmatching orbits satisfying the same trade condition, one is the coalescence parameter, and only the matching point splits completely into linear factors. The exceptional one-factorizations are classical; the boundary we isolate is that fixed line and its unique completely split point.

The 14- and 22-point homogenizations are self-associated and arithmetically Gorenstein, with the cubic as the Macaulay inverse system of an Artinian reduction—a consequence of general self-dual-code criteria for the Schur square, not a hypothesis.

**Keywords.** conic ideal; secant matching; strength-two trade; one-factorization; Dickson polynomial; self-associated points; arithmetically Gorenstein; cubic inverse system.

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## 1 Introduction

A perfect matching of marked conic points determines a product of secant lines. Restriction to the conic deliberately forgets the pairing, and we ask when low-degree algebra in the resulting quotient recovers it. Over all odd finite fields, a one-dimensional two-valued quadratic trade is rigid enough: within the full matching carrier it forces exactly the $B_3/\mathbb{F}_7$ and $H_3/\mathbb{F}_{11}$ orbits, and the first surviving odd tensor restores the orientation of the recovered sheets. The Clebsch-related $H_3/\mathbb{F}_{11}$ case is one exceptional output, not an input.

A four-endpoint switch shows that competing secant products differ by the conic equation, so their quotients form an affine configuration. For the two surviving orbits, quadratic products recover an unordered bipartition of that configuration. The first surviving signed tensor is cubic and orients its two parts. Thus quadratic data recover the factorization without choosing an orientation; the first odd covariant restores the choice. By the *matching carrier* we mean the locus of products of the secant lines in a perfect matching of the marked conic points. This is a low-degree recognition theorem rather than an orbit census: within the full matching carrier, the quadratic relation forces one of two exceptional geometries without assuming its group type, self-association, or Gorenstein coordinate ring.

The general implication from self-dual codes and their Schur squares to Gorenstein coordinate rings is background here. The theorem instead starts from the weaker quadratic-trade condition, with no self-duality or Gorenstein premise.

| type | linear quotient | sheets/cubic |
|---|---|---|
| $A_3/\mathbb{F}_5$ | rank 3 | not applicable |
| $B_3/\mathbb{F}_7$ | rank 6 | recovered/oriented |
| $H_3/\mathbb{F}_{11}$ | rank 10 | recovered/oriented |

The all-field bottleneck is *outer parity*. A selected simple $H = \mathrm{PSL}_2(q)$-detector has two $G = \mathrm{PGL}_2(q)$-extensions, exchanged by the determinant square-class character; the trade forces the opposite one into quadratic moments. Appendix A rules it out, Faber makes the detector cases exhaustive, and Dickson enters only after the sheet size is known.

**Theorem 1.1** (Quadratic-trade classification and cubic orientation)**.** *Let the $A_3$, $B_3$, and $H_3$ matching configurations be the finite matching orbits defined in Section 3. Then:*

(i) *among all odd prime powers $q$, the $B_3/\mathbb{F}_7$ and $H_3/\mathbb{F}_{11}$ configurations are the only full perfect-matching orbits under $\mathrm{PGL}_2(q)$ whose strength-two trade space is one-dimensional and generated by a vector with exactly two values;*

(ii) *matching products scaled as in Section 2 have equal restriction to the conic, and their conic-ideal quotients have difference ranks* 3, 6, *and* 10*;*

(iii) *in the $B_3$ and $H_3$ cases, the two factorization sheets are, up to interchange, the unique complementary halves with equal first and second tensor moments;*

(iv) *in those two cases, the signed moments vanish through degree two, while a nonzero degree-three tensor transforms by the character that exchanges the sheets;*

(v) *the homogenized $B_3$ and $H_3$ configurations are self-associated with Gorenstein coordinate rings; the signed cubic is the Macaulay inverse system of their Artinian reductions;*

(vi) *for either surviving stabilizer, the fixed locus in the ambient conic fiber is an affine line whose unique secant-product point is the matching point, while $q-2$ nonmatching orbits on that line still have one-dimensional two-valued quadratic-trade space.*

Rodríguez-Pajares, Ruano, and Salizzoni characterize the Gorenstein defect of a self-dual code by its indecomposable blocks, equivalently by the defect of its Schur square [13, Theorem 3.11 and Corollaries 3.12–3.13]. That general mechanism is not a priority claim here. The new input and outputs are the all-field trade classification, sheet reconstruction, sharp matching boundary, and orbit-specific cubic orientation in the theorem above.

**Proof and reading map.**

Matching products have a common conic restriction; differences define the conic-ideal quotient.
*Section 2*

$\swarrow$ $\searrow$

Compute the three quotient ranks $3, 6, 10$.
*Section 3; (ii)*

$\Downarrow$
rank data used below

A two-valued quadratic trade forces two $\mathrm{PSL}_2(q)$-sheets.
*Section 3*

$\Downarrow$

Faber's subgroup split is exhaustive; the detectors force each sheet to have size $q$.
*Section 3; Appendix A*

$\Downarrow$

Dickson's classification and the invariant-matching check identify the only possible survivors as $B_3/\mathbb{F}_7$ and $H_3/\mathbb{F}_{11}$.
*Section 3; (i), necessity*

$\searrow$ $\swarrow$

The moment identities prove the converse in (i), recover the unordered sheets, and show that the signed cubic orients them and gives Gorenstein duality.
*Section 4; (i), converse; (iii)–(v)*

$\Downarrow$

The fixed affine line exposes the off-carrier ambiguity; its inherited trade and geometric splitting give the sharp carrier boundary.
*Section 4; (vi)*

The right-hand chain proves the necessity in (i); the two branches merge for its converse and the recovery/orientation results. The boundary inherits its trade clause from that merged node. Appendix A is required only at the detector node. Appendices B–F may be skipped on a first pass; Appendix G records the trust boundary and supplies no logical premise.

For each surviving stabilizer, Section 4 constructs a fixed affine line inside the ambient conic fiber. The forms in that ambient fiber that split geometrically into linear factors comprise its geometric Chow locus. The carrier boundary is summarized by

$$\begin{aligned} \text{matching carrier and two-valued trade} \quad &\Longrightarrow \quad B_3/\mathbb{F}_7 \text{ or } H_3/\mathbb{F}_{11}, \\ \text{fixed conic-fiber line away from coalescence} \quad &\Longrightarrow \quad q-1 \text{ trade-condition orbits}, \\ \text{fixed conic-fiber line} \cap \text{geometric Chow locus} \quad &= \quad \{\text{matching point}\}. \end{aligned} \tag{1.1}$$

The middle line consists of one matching orbit and $q-2$ nonmatching orbits; the excluded rational point is the coalescence parameter. The last line shows that imposing the Chow condition restores uniqueness.

## 2 Conic matching products

Write the standard conic as

$$\mathcal{C} = V(Q), \qquad Q = XZ - Y^2,$$

and parametrize it by $\nu(s:t) = (s^2 : st : t^2)$. For points $a = (a_0 : a_1)$ and $b = (b_0 : b_1)$ of $\mathbb{P}^1$, put $[a,b] = a_0b_1 - a_1b_0$. We scale the secant $L_{ab}$ through $\nu(a)$ and $\nu(b)$ by the condition

$$L_{ab}(\nu(s:t)) = [a,(s:t)]\,[b,(s:t)].$$

Let $\Omega$ be a set of $2m$ distinct marked points of $\mathbb{P}^1$, with fixed vector representatives. For a perfect matching $M$ of $\Omega$, define its matching product

$$P_M = \prod_{\{a,b\}\in M} L_{ab}.$$

The vector representatives are part of the marked-conic data. Their role is to make the products comparable before passing to projective equivalence.

This choice does not add projective structure. If the representative of each endpoint $a$ is rescaled by $\lambda_a$, then every matching product is multiplied by the same scalar $\prod_{a\in\Omega}\lambda_a$. Hence every quotient difference in (2.1) is multiplied by that common scalar: the affine configuration changes by a homothety, while its trade data, ranks, and cubic line are unchanged projectively.

**Proposition 2.1** (The matching-secant quotient)**.** *For every $m \geq 2$ and every two perfect matchings $M, N$ of $\Omega$,*

$$P_M|_{\mathcal{C}} = P_N|_{\mathcal{C}}, \qquad P_M - P_N \in (Q).$$

*Consequently, after choosing a reference matching $M_0$,*

$$\Phi_{M_0}(M) = \frac{P_M - P_{M_0}}{Q} \in H^0(\mathbb{P}^2, \mathcal{O}(m-2)) \tag{2.1}$$

*is an affine configuration. Changing the reference matching translates this configuration. If $g \in \mathrm{GL}_2$ preserves $\Omega$, $\rho(g)$ is its action on binary quadratics, and representatives are transported by $g$, then*

$$\Phi_{gM_0}(gM)\circ\rho(g) = \det(g)^{2m-2}\Phi_{M_0}(M). \tag{2.2}$$

*Thus the difference space, its rank, and its affine moment conditions do not depend on $M_0$ and are equivariant under the endpoint stabilizer.*

*Proof.* For $x \in \mathbb{P}^1$,

$$P_M(\nu(x)) = \prod_{a\in\Omega}[a,x],$$

which does not depend on the pairing. The kernel of the Veronese pullback

$$k[X,Y,Z] \longrightarrow k[s,t], \qquad (X,Y,Z) \longmapsto (s^2, st, t^2)$$

is the principal ideal $(Q)$, so $Q$ divides every difference. Changing the reference follows by subtraction. Finally, $L_{ga,gb}\circ\rho(g) = \det(g)^2 L_{ab}$ and $Q\circ\rho(g) = \det(g)^2 Q$, which gives (2.2). □

## 2.1 The four-endpoint switch

For four distinct endpoints, the Plücker relation gives the local switch identity

$$L_{ab}L_{cd} - L_{ac}L_{bd} = [a,d][b,c]\,Q. \tag{2.3}$$

Multiplying (2.3) by the secant factors common to two matchings proves divisibility for a single switch. Any two perfect matchings are joined by a sequence of such switches: if $M$ and $N$ differ at $a$, write $\{a,b\} \in M$ and $\{a,c\} \in N$; switching the two $M$-edges through $b$ and $c$ creates $\{a,c\}$ and reduces the symmetric difference. This also gives an elementary proof of the divisibility clause in Proposition 2.1.

The switch calculation and the global pullback proof play different roles. The pullback proves the quotient for arbitrary matchings at once. The switch identity records how local changes of factorization move in the quotient space. A selected Coxeter orbit need not contain every perfect matching or every intermediate switch.

# 3 Matching-orbit classification and quotient ranks

The three distinguished finite matching orbits are discrete input: Proposition 2.1 applies to every matching, but it does not distinguish the Coxeter orbits. Here and below, "rank three" refers to Coxeter rank. It does not refer to the dimensions of the conic-ideal quotient spaces, which are $3, 6, 10$.

For $q \in \{5, 7, 11\}$, index

$$\mathbb{P}^1(\mathbb{F}_q) = \{0, 1, \ldots, q-1, \infty\},$$

where $a$ denotes $(1 : a)$ and $\infty$ denotes $(0 : 1)$. Let $G_q = \mathrm{PGL}_2(q)$ act in the usual way. The three base matchings are

| $T$ | $q$ | $M_T$ |
|---|---|---|
| $A_3$ | 5 | $\{0,\infty\}, \{1,4\}, \{2,3\}$, |
| $B_3$ | 7 | $\{0,2\}, \{1,4\}, \{3,\infty\}, \{5,6\}$, |
| $H_3$ | 11 | $\{0,1\}, \{2,5\}, \{3,7\}, \{4,9\}, \{6,8\}, \{10,\infty\}$. |

(3.1)

Define the *rank-three matching configuration*

$$\Omega_T = G_q \cdot M_T.$$

The stabilizers of the three displayed matchings are respectively $S_4, S_4, A_5$. For a direct check, with $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ acting by $x \mapsto (ax+b)/(cx+d)$, generators preserving the three rows of (3.1) may be chosen as

| $T$ | $a$ | $b$ | relations |
|---|---|---|---|
| $A_3$ | $\left(\begin{smallmatrix} 0 & 1 \\ 2 & 0 \end{smallmatrix}\right)$ | $\left(\begin{smallmatrix} 1 & 1 \\ 2 & 3 \end{smallmatrix}\right)$ | $a^2 = b^3 = (ab)^4 = 1$ |
| $B_3$ | $\left(\begin{smallmatrix} 0 & 1 \\ 3 & 2 \end{smallmatrix}\right)$ | $\left(\begin{smallmatrix} 1 & 1 \\ 1 & 5 \end{smallmatrix}\right)$ | $a^3 = b^4 = (ab)^2 = 1$ |
| $H_3$ | $\left(\begin{smallmatrix} 0 & 1 \\ 2 & 3 \end{smallmatrix}\right)$ | $\left(\begin{smallmatrix} 0 & 1 \\ 7 & 8 \end{smallmatrix}\right)$ | $a^3 = b^5 = (ab)^2 = 1.$ |

These are the standard triangle presentations of $S_4, S_4, A_5$; the maximal-subgroup classification and the outer-normalizer check [7] show that they are the full stabilizers in $G_q$. Thus orbit–stabilizer gives

$$|\Omega_{A_3}| = 5, \qquad |\Omega_{B_3}| = 14, \qquad |\Omega_{H_3}| = 22. \tag{3.2}$$

These matching orbits and their stabilizers are classical. In particular, Edge and Dye identify the exceptional finite conic configurations and substantial parts of their incidence geometry [2, 3]. The exceptional one-factorizations themselves also precede this work: Cameron and Korchmáros classify the doubly transitive case, and Han's classification of symmetric factorizations contains the triples $\mathrm{PSL}_2(7), S_4, 8, 7$ and $\mathrm{PSL}_2(11), A_5, 12, 11$ [4, 6]. We use the Coxeter names for the corresponding tetrahedral, octahedral, and icosahedral data; neither the orbit counts in (3.2) nor the ambiguity of an unmarked conic is asserted here as new.

Two adjacent frameworks should also be separated from the claim here. Korchmáros, Pace, and Sonnino construct one-factorizations from partitions of the external points of an oval, each part meeting every tangent once [5]; the present carrier instead consists of secant matchings on the marked conic, followed by passage through its ideal. Likewise, the group–normal-subgroup architecture in which one group fixes the factors and a larger group permutes them is the classical homogeneous factorization framework of Giudici, Li, Potočnik, and Praeger [8]. The assertions below concern the specific conic-quotient and trade-rigidity mechanism, not that architecture.

### 3.1 Trade reduction and the modular obstruction

Here is the application dictionary for the abstract reduction. We take $G = \mathrm{PGL}_2(q)$, $H = \mathrm{PSL}_2(q)$, and $\Omega$ to be a full matching orbit; once the two trade levels are identified with its $H$-orbits, they are the candidate sheets. The module $\mathcal{A}$ is the space of ambient affine-linear functions, $E = \mathcal{A}^\vee$ is its homogenized point-vector module, and $F = \ker(E \to k) = \mathrm{Sym}^{(q-3)/2} L(2)$ is the translation module. The quadratic moment map is $\mu([M]) = v_M^2$. The next lemma converts the trade relation into simultaneous occurrences of the two $G$-extensions of one $H$-detector in $\mathrm{Sym}^2 E$; the uniform exclusion later rules out the opposite-parity occurrence by eliminating both its zero-linear and split-pullback alternatives.

**Lemma 3.1** (Projective–trade reduction)**.** *Let $k$ be an algebraically closed field of odd characteristic, let $H \lhd G$ have index two, and let a transitive $G$-set $\Omega$ split as two transitive $H$-sets $\Omega_+ \sqcup \Omega_-$. Let*

$$0 \longrightarrow F \longrightarrow E \xrightarrow{\epsilon} k \longrightarrow 0$$

*be an exact sequence of $kG$-modules, and let $v : \Omega \to E$ be a $G$-map with $\epsilon(v_\omega) = 1$. Define*

$$\mu : k[\Omega] \longrightarrow \mathrm{Sym}^2 E, \qquad [\omega] \longmapsto v_\omega^2.$$

*Also put*

$$a : k[\Omega] \longrightarrow E, \qquad [\omega] \longmapsto v_\omega.$$

*All embeddings, pullbacks, and splittings below are in the category of $kG$-modules. Assume that*

$$\ker \mu = k \left( \sum_{\omega \in \Omega_+} [\omega] - \sum_{\omega \in \Omega_-} [\omega] \right),$$

*and that $k[\Omega_+]$ contains a nontrivial simple $H$-submodule $S$ such that ${}^g S \simeq S$ for $g \in G \setminus H$, and that $S$ extends to $G$. If $S^+$ and $S^- = S^+ \otimes \chi$ are its two $G$-extensions, where $\chi$ is the nontrivial character of $G/H$, then both $S^+$ and $S^-$ embed in $\mathrm{Sym}^2 E$. For either summand $T = S^\pm$, let $j = \mu|_T : T \to \mathrm{Sym}^2 E$ be the resulting embedding and put $i = \partial j$. Exactly one of the following occurs:*

1. *$i = 0$ and $j(T) \subseteq \mathrm{Sym}^2 F$;*

*2. $i : T \to E$ is an embedding and the pullback along $i$ of*

$$0 \longrightarrow \operatorname{Sym}^2 F \longrightarrow \operatorname{Sym}^2 E \xrightarrow{\partial} E \longrightarrow 0$$

*splits, where*

$$\partial(xy) = \epsilon(x)y + \epsilon(y)x.$$

*Moreover, let $\xi \in \operatorname{Ext}^1_G(k, F)$ be the class of the affine extension and let $\pi : F \otimes F \to \operatorname{Sym}^2 F$ be the canonical quotient, $\pi(f \otimes f') = ff'$. Since $T$ is nontrivial, $i(T) \subseteq F$, and*

$$\delta_T(i) = i^* \pi_*(F \otimes \xi) \quad \text{in} \quad \operatorname{Ext}^1_G(T, \operatorname{Sym}^2 F). \tag{3.2a}$$

*Proof.* Choose $g \in G \setminus H$. The $G$-submodule of $k[\Omega]$ generated by $S$ is $S \oplus {}^g S$. It has zero intersection with the displayed kernel, since that kernel is trivial as an $H$-module and $S$ is not. Thus $\mu$ embeds $S \oplus {}^g S$.

Choose the extension $S^+$. The elementary tensor identity for induction gives

$$\operatorname{Ind}_H^G S \xrightarrow{\sim} S \oplus {}^g S \simeq S^+ \otimes k[G/H].$$

The first map sends $x \otimes s$ to $xs$; it is an isomorphism because its two sheet-supported images are disjoint. Since $2 \neq 0$ in $k$, one has $k[G/H] \simeq k \oplus k_\chi$. Hence the last module is $S^+ \oplus S^-$, proving that both parities embed.

It remains only to inspect the canonical map $\partial$. It is $G$-equivariant, and on every basis vector

$$\partial\mu([\omega]) = 2v_\omega = 2a([\omega]).$$

Choose $e \in E$ with $\epsilon(e) = 1$, merely as a vector-space splitting. Then

$$\operatorname{Sym}^2 E = ke^2 \oplus eF \oplus \operatorname{Sym}^2 F, \qquad \partial(e^2) = 2e, \quad \partial(ef) = f.$$

Consequently the displayed sequence is exact. Since $T$ is simple, $i$ is either zero or injective. In the first case the image of $j$ lies in $\operatorname{Sym}^2 F$. In the second, the map $s \mapsto (j(s), s)$ is a section of the pullback of that sequence along $i$. Equivalently, if

$$\delta_T : \operatorname{Hom}_G(T, E) \longrightarrow \operatorname{Ext}^1_G(T, \operatorname{Sym}^2 F)$$

is the connecting map, then $\delta_T(i) = 0$. This proves the dichotomy. We use the convention that $\delta_T(i)$ is the Yoneda class of the pullback along $i$.

For the final assertion, put $D = \partial^{-1}(F)$. Multiplication $f \otimes x \mapsto fx$ gives a morphism from

$$0 \longrightarrow F \otimes F \longrightarrow F \otimes E \xrightarrow{1 \otimes \epsilon} F \longrightarrow 0$$

onto

$$0 \longrightarrow \operatorname{Sym}^2 F \longrightarrow D \xrightarrow{\partial} F \longrightarrow 0.$$

The left map is $\pi$, the right map is the identity, and the lower row is the pushout of the upper row: this is immediate from the vector-space splitting $E = ke \oplus F$ used above. Thus the lower row represents $\pi_*(F \otimes \xi)$. Since $\epsilon i : T \to k$ is zero for nontrivial simple $T$, the map $i$ lands in $F$; pulling back the lower row gives (3.2a). More concretely, choose the same vector-space lift $e$ and put $c(g) = ge - e \in F$. With the connecting-cocycle convention $g \mapsto gsg^{-1} - s$, the class in (3.2a) is represented by

$$z_i(g) : T \longrightarrow \operatorname{Sym}^2 F, \qquad z_i(g)(t) = i(t)c(g). \tag{3.2b}$$

Changing $e$ changes $z_i$ by a coboundary. □

## 3.2 Detector modules and their outer parity

The all-field exclusion needs no classification of the full finite-group socle of $F$. It uses only two extension-field linear channels, the prime-field Fischer summands, and the absence of one outer parity from the quadratic moment module. We state exactly those interfaces here; their specialist proofs are collected in Appendix A.

**Lemma 3.2** (Targeted linear detectors)**.** *Let $q = p^e$ be odd, let $H = \mathrm{PSL}_2(q)$, and put*

$$d = \frac{q-3}{2}, \qquad F = \mathrm{Sym}^d L(2).$$

1. *For the Steinberg module $\mathrm{St} = L(q-1)$,*

$$\mathrm{Hom}_H(\mathrm{St}, F) = 0.$$

2. *If $q \equiv 3 \pmod 4$, $e > 1$, and $S = L(q-7)$, then*

$$S \hookrightarrow F.$$

   *Every $H$-map in this channel is algebraic. In the covariant polynomial convention the displayed map is $L(q-7) \otimes \det^2 \to F$; after normalizing the target by $\det^{-d}$, its source twist is $\det^{2-d} = \det^{-(q-7)/2}$. Thus every occurrence has this determinant-normalized outer parity, and the other $\mathrm{PGL}_2(q)$-extension is absent from $F$.*

3. *If $e = 1$ and $p \geq 5$, then*

$$F = \bigoplus_{0 \leq j \leq \lfloor (p-3)/4 \rfloor} L(p-3-4j),$$

   *with every summand occurring once and admitting an $H$-retraction.*

*Proof.* See Appendix A.1. The root-degree bound is $q-1$ in the two extension-field channels, so their finite maps are algebraic. The $L(q-7)$ map is multiplication of a digitwise Cartan embedding by the discriminant, and the prime-field statement follows from the tilting Fischer character identity. □

**Lemma 3.3** (Opposite-parity quadratic obstruction)**.** *Let $S$ be one of the detector modules used below: the Steinberg module, $L(q-7)$ in the extension-field branch, or one of the prime-field modules $L(s)$, $s \in \{2, 4, 6, 8, 12\}$, in the range displayed in (3.2f). Let $S^\square$ be the extension to $G = \mathrm{PGL}_2(q)$ opposite to the covariant determinant-normalized extension. Then*

$$\mathrm{Hom}_G(S^\square, \mathrm{Sym}^2 F) = 0,$$

*and the other extension is $S^\square \otimes \chi$, where $\chi$ is the nontrivial character of $G/H$.*

*Proof.* Appendix A.2 constructs an injection from this finite-group Hom space into

$$\mathrm{Hom}_{\mathrm{SL}_2(k)}(S \otimes L(1) \otimes L(1)^{(e)}, \mathrm{Sym}^2 F)$$

and proves the latter space zero for each listed source. The Steinberg case uses the complete two-section Weyl filtration of $T(2q)$, not only its head. □

**Lemma 3.4** (Affine-class contraction). *Let $i : S^{\square} \hookrightarrow F$ have an $H$-retraction $\rho : F \to S$. For the point-vector extension $0 \to F \to E \to k \to 0$, with class $[c]$, the pullback along $i$ of*

$$0 \longrightarrow \operatorname{Sym}^2 F \longrightarrow \operatorname{Sym}^2 E \xrightarrow{\partial} E \longrightarrow 0$$

*contracts to*

$$(\dim S)[c] + i_* \rho_* [c] \in H^1(H, F). \tag{3.2c}$$

*For the prime-field Fischer detectors, $[c] \neq 0$, it is supported only on the top summand $L(p-3)$, and the resulting scalar in (3.2c) is nonzero.*

*Proof.* The evaluation–coevaluation calculation, the polynomial representative of $[c]$, and the Borel-cohomology support calculation are given in Appendix A.3. $\square$

### 3.3 Uniform sheet exclusion

**Lemma 3.5** (Uniform nonprincipal-sheet exclusion). *Let $q = p^e$ be odd, let $G = \mathrm{PGL}_2(q)$ and $G^+ = \mathrm{PSL}_2(q)$, and let $\Omega$ be a full $G$-orbit of perfect matchings split into two $G^+$-orbits. Let $L \subseteq \mathbb{F}_q^{\Omega}$ be its affine evaluation space. If $(L^{\circ 2})^{\perp} \subseteq \mathbb{F}_q^{\Omega}$ is the sheet-sign line, then either sheet has $q$ members.*

*Proof.* We give the modular argument because it removes the one-factorization hypothesis. Let $K \leq G^+$ stabilize a matching and write a sheet as $G^+/K$. A nontrivial unipotent cannot fix a perfect matching: it has one fixed endpoint, while the partner of that endpoint would have an orbit of length $p$. Thus $K$ is a $p'$-group and

$$P_0 = \mathbb{F}_q[G^+/K]$$

is projective. Its restriction to a translation Sylow subgroup is a sum of regular modules, and

$$|G^+/K| = q\lambda \tag{3.2c$_0$}$$

for some positive integer $\lambda$. Extend scalars to $k = \overline{\mathbb{F}}_q$ and put $P = k \otimes_{\mathbb{F}_q} P_0$.

Let $\mathcal{A}$ be the full space of affine-linear functions on the ambient conic-quotient form space, after scalar extension to $k$, and let $\mathcal{A} \twoheadrightarrow L_k$ be restriction to $\Omega$. Put $E = \mathcal{A}^*$. Thus $E$ is the ambient point-vector module; it is not the dual of the possibly smaller orbit span $L_k$. The inclusion of the constants in $\mathcal{A}$ dualizes to

$$0 \longrightarrow F \longrightarrow E \xrightarrow{\epsilon} k \longrightarrow 0, \qquad F = \operatorname{Sym}^{(q-3)/2} L(2), \tag{3.2d}$$

and evaluation at a matching $M$ is a vector $v_M \in E$ with $\epsilon(v_M) = 1$. The evaluation of products of functions in $\mathcal{A}$ factors as

$$\operatorname{Sym}^2 \mathcal{A} \twoheadrightarrow \operatorname{Sym}^2 L_k \longrightarrow k^{\Omega}.$$

Its transpose is the moment map in Lemma 3.1. Since the first arrow is surjective, the kernel of that transpose is exactly the annihilator of $L_k^{\circ 2}$, hence the scalar extension of the strength-two trade space. Here we use that scalar extension is flat and commutes with $\operatorname{Sym}^2$. This ambient formulation is essential when the orbit spans a proper subspace, as it does for $H_3$. Write $\mu$ for this moment map and $a : k[\Omega] \to E$ for the linear moment map $[M] \mapsto v_M$.

We apply Lemmas 3.2, 3.3, and 3.4. Denote the absent outer extension by $S^{\square}$, and write

$$j_{\square} = \mu|_{S^{\square}}, \qquad i_{\square} = \partial j_{\square} = 2a|_{S^{\square}}.$$

If $i_\square = 0$, then $j_\square$ would embed $S^\square$ in $\mathrm{Sym}^2 F$, contradicting Lemma 3.3. We choose $S$ so that this already occurs over extension fields. Over prime fields, a selected linear copy instead retracts and Lemma 3.4 rules out the split pullback.

The list of possible stabilizers is exhaustive without a recursive subfield argument. Since $K$ is $p'$, it is a finite $p$-regular subgroup of $\mathrm{PGL}_2(\overline{\mathbb{F}}_p)$. Faber's tame subgroup theorem [9, Theorem C] gives

$$K \text{ cyclic, dihedral, } A_4, S_4, \text{ or } A_5. \tag{3.2e}$$

Only this abstract type is used in the following detector split.

Suppose first that $e > 1$. If a cyclic or dihedral $K$ is intransitive on $\mathbb{P}^1(\mathbb{F}_q)$, take $S = \mathrm{St} = L(q-1)$. The endpoint permutation module is

$$k[\mathbb{P}^1(\mathbb{F}_q)] = k \oplus \mathrm{St},$$

so $\dim S^K = |K\backslash\mathbb{P}^1(\mathbb{F}_q)| - 1 > 0$. Lemma 3.2 gives $\mathrm{Hom}_H(S, F) = 0$; hence $i_\square = 0$, a contradiction.

It remains among cyclic and dihedral groups only to consider a transitive dihedral $K$. A cyclic subgroup of $H$ lies in a nonsplit torus of order at most $(q+1)/2$, so it cannot be transitive on $q+1$ endpoints. Thus $K$ is the full nonsplit normalizer of order $q+1$, acting regularly. Put $n = (q+1)/2$, choose a generator $z$ of the nonsplit torus of order $2n$ in $G$, with $z \notin G^+$, and write

$$r = z^2, \qquad K = \langle r, w : r^n = w^2 = 1,\ wrw = r^{-1}\rangle.$$

A $K$-invariant matching on the regular $K$-set is right multiplication by a nonidentity involution. Since $zwz^{-1} = rw$, conjugation by $r^a z$ sends $r^j w$ to $r^{j+2a+1}w$, preserving the matching exactly when

$$2a + 1 = 0 \pmod{n}. \tag{3.2f$_0$}$$

If $n$ is even, the rotation involution $r^{n/2}$ is fixed by $z$; if $n$ is odd, (3.2f$_0$) is soluble for every reflection matching. Thus a split dihedral orbit can only be a reflection matching with $n$ even. Hence $q \equiv 3 \pmod 4$, and we take $S = L(q-7)$. Its unique zero-weight line has Weyl sign $+1$, so $S^K \neq 0$. Lemma 3.2 puts every linear occurrence in the determinant-normalized extension; therefore $a|_{S^\square} = 0$, again a contradiction.

If $K$ is exceptional and intransitive, the same Steinberg argument applies:

$$\dim \mathrm{St}^K = |K\backslash\mathbb{P}^1(\mathbb{F}_q)| - 1 > 0.$$

If it is transitive, then $q+1 \mid |K|$, while $|K| \in \{12, 24, 60\}$. For an odd nonprime prime power this leaves only $q = 9$, where $p = 3$ divides every exceptional order, contrary to the $p'$ hypothesis. This closes the extension-field cases.

Now let $e = 1$. For cyclic, dihedral, and exceptional stabilizers use

| type and range | $S$ |
|---|---|
| cyclic, $p \geq 5$ | $L(2)$ |
| dihedral, $p \geq 7$ | $L(4)$ |
| $A_4$, $p \geq 11$ | $L(6)$ |
| $S_4$, $p \geq 11$ | $L(8)$ |
| $A_5$, $p \geq 17$ | $L(12)$. |

(3.2f)

The cyclic and dihedral modules have the required zero-weight line, with Weyl signs $-1$ and $+1$. For the exceptional groups, averaging the ordinary symmetric-power character over the binary-polyhedral lift gives

| $K$ | $S$ | $\lvert K\rvert \dim S^K$ |
|---|---|---|
| $A_4$ | $L(6)$ | 12 |
| $S_4$ | $L(8)$ | 24 |
| $A_5$ | $L(12)$ | 60. |

Here the (class size, lift order) pairs are respectively

$$\begin{array}{c|l} A_4 & (1,1),(3,4),(8,6) \\ S_4 & (1,1),(9,4),(8,6),(6,8) \\ A_5 & (1,1),(15,4),(20,6),(12,10),(12,5). \end{array}$$

Thus $\dim S^K = 1$ in every exceptional row. Since $K$ is $p'$, averaging is valid in $k$. Frobenius reciprocity and the self-dual permutation form embed each self-dual $S$ in $P = k[G^+/K]$.

By Lemma 3.2, each selected $S$ is either absent from $F$, giving $i_\square = 0$, or is a multiplicity-one Fischer summand. In the latter case $i_\square = 0$ gives the same immediate contradiction; otherwise $i_\square$ identifies the unique copy and its Fischer projection is an $H$-retraction. Lemma 3.4 then makes the pulled-back affine class nonzero. Lemma 3.3 excludes the alternative quadratic embedding in both cases.

The case tree is now summarized by the detector or obstruction used in each branch:

| regime | stabilizer | detector or obstruction |
|---|---|---|
| $e > 1$ | intransitive cyclic, dihedral, exceptional | St, zero linear moment |
| $e > 1$ | full nonsplit normalizer | $L(q-7)$, opposite linear parity |
| $e > 1$ | transitive exceptional | only $q = 9$, but then $p \mid \lvert K\rvert$ |
| $e = 1$ | cyclic, dihedral, $A_4, S_4, A_5$ | (3.2f), then absence or contraction |
| small | $q = 3, 5, 7, 13$ | one orbit, outer stabilizer, or no subgroup |

The remaining endpoint statements make the last row explicit:

| $q$ | $K$ | orbits | unique matching data | $N_G(K)$ |
|---|---|---|---|---|
| 5 | $A_4$ | 6 | $C_2 < V_4$ is the two-point block | $S_4$ |
| 7 | $A_4$ | $4+4$ | $N_{A_4}(C_3) = C_3$ gives the cross-orbit pairing | $S_4$ |
| 5 | $D_6$ | 6 | reflection, $n = 3$ in (3.2f$_0$) | $D_{12}$. |

(3.2g$_0$)

In each row the displayed outer normalizer preserves the unique matching, so the full orbit does not give a split $\lambda > 1$ case. For $q = 3$, the three perfect matchings form one $G^+$-orbit, so the two-sheet premise is impossible. There is no $A_5$ row at $q = 13$, since $60 \nmid \lvert \mathrm{PSL}_2(13)\rvert$. We have therefore produced the quadratic obstruction for every possible split stabilizer with $\lambda > 1$, a contradiction. Hence $\lambda = 1$. $\square$

### 3.4 Balanced-orbit completeness

**Theorem 3.6** (Uniform balanced-orbit classification)**.** *Let $q$ be an odd prime power, put*

$$G = \mathrm{PGL}_2(q), \qquad G^+ = \mathrm{PSL}_2(q),$$

*and let $\Omega$ be a $G$-orbit of perfect matchings of $\mathbb{P}^1(\mathbb{F}_q)$. Form its conic-ideal quotient configuration and let $L \subseteq \mathbb{F}_q^\Omega$ be its affine evaluation space. Put*

$$L^{\circ 2} = \operatorname{span}\{fg : f, g \in L\},$$

*with pointwise multiplication, and take its annihilator under the standard coordinate pairing on $\mathbb{F}_q^\Omega$. Suppose $(L^{\circ 2})^\perp$ is one-dimensional and every nonzero vector in it takes exactly two values. Then*

$$(q,\Omega) = (7, \Omega_{B_3}) \quad \text{or} \quad (q,\Omega) = (11, \Omega_{H_3}). \tag{3.2h}$$

*Conversely, both displayed orbits satisfy these intrinsic hypotheses. For either orbit, the two level sets are its two one-factorization sheets.*

*Proof.* Proposition 2.1 makes $L$, its square, and the line

$$R = (L^{\circ 2})^{\perp}$$

$G$-stable. Choose $0 \neq \tau \in R$. Its zero set is $G$-invariant, so transitivity gives $\tau$ full support. The two level sets of $\tau$ depend only on $R$, hence $G$ permutes them. This action is nontrivial because $G$ is transitive on $\Omega$.

For $q > 3$, the commutator subgroup of $G$ is $G^+$, and $G/G^+$ has order two. The kernel of the action on the two levels is therefore $G^+$. The stabilizer of a point fixes its level and so lies in $G^+$; consequently the two levels are exactly the two $G^+$-orbits. An outer element acts on $R$ by a scalar and exchanges the two nonzero values of $\tau$, so those values are opposites.

Lemma 3.5 shows that each intrinsic level has size $q$. Thus $|\Omega| = 2q$. When $q = 3$, there are only three perfect matchings altogether, so such an orbit cannot occur.

Now fix $M \in \Omega$ and let $K = \mathrm{Stab}_G(M)$. Orbit–stabilizer and the derived two-sheet conclusion give

$$|K| = \frac{q^2 - 1}{2}, \qquad K \leq G^+, \qquad [G^+ : K] = q. \tag{3.2i}$$

Assume $q \geq 5$, and place $K$ in a maximal subgroup of $G^+$. Dickson's maximal-subgroup theorem, in the form recorded by Giudici [7, Theorem 2.2], leaves Borel, dihedral, subfield, and exceptional $A_4, S_4, A_5$ subgroups. The first two have order smaller than $(q^2 - 1)/2$. A proper subfield subgroup is no larger than $\mathrm{PGL}_2(r)$, where $q = r^e$ and $e \geq 2$, and

$$r(r^2 - 1) < \frac{r^{2e} - 1}{2}.$$

Thus it too is too small. An exceptional maximal subgroup has order at most 60, so (3.2i) leaves $q = 5, 7, 9, 11$. For $q = 9$, the required order 40 divides none of $12, 24, 60$. In the other three cases, order and divisibility force

$$(q, K) = (5, A_4),\ (7, S_4),\ (11, A_5). \tag{3.2j}$$

It remains to ask which of these subgroup actions preserves a matching. This is a block-system question, not a matching census. For $q = 5$, the $A_4$-action on the six endpoints is transitive with point stabilizer $C_2$. Its normal Klein four subgroup gives the unique two-point block system. The normalizer of $A_4$ in $\mathrm{PGL}_2(5)$ is $S_4$; uniqueness makes this whole normalizer preserve the matching. Its full orbit therefore has five points, namely $\Omega_{A_3}$, rather than ten. In particular, the other ten matchings cannot form a balanced $5+5$ orbit.

For $q = 7$, the $S_4$-action on eight endpoints has point stabilizer $C_3$. A two-point block stabilizer must have order six, and the unique possibility containing that $C_3$ is its normalizer $S_3$. Hence the invariant matching is unique. Similarly, for $q = 11$, the $A_5$-action on twelve endpoints has point stabilizer $C_5$; its unique possible two-point block stabilizer is $N_{A_5}(C_5) = D_{10}$. The invariant matching is again unique. In each case the two $G^+$-classes of the exceptional subgroup are fused by $G$, and the subgroup has no outer normalizer [7, Lemma 2.3]: the diagonal outer automorphism fuses the two classes, so no element in its coset normalizes a representative. Indeed, if $J = \mathrm{Stab}_G(M)$, then $J \cap G^+$ contains $K$. Maximality gives $J \cap G^+ = K$ or $G^+$, and the latter is impossible because $G^+$ does not fix $M$. If $J > K$, an element of $J \setminus K$ would therefore be an outer normalizer of $K$, contrary to the cited fusion statement. Thus $J = K$, so the orbit has size $2q$ and splits into the two claimed $G^+$-orbits. The unique block systems are precisely the matchings displayed in (3.1), up to $G$-conjugacy.

Conversely, $q = 7, 11$ are both 3 (mod 4), so $G^+$ is transitive on unordered pairs of endpoints. Each $q$-matching sheet therefore covers every edge equally often; its total of $q(q+1)/2$ edge occurrences equals the number of edges, so it is a one-factorization. Propositions 4.1 and 4.2 identify $(L^{\circ 2})^{\perp}$ with the sheet-sign line. Hence both displayed orbits satisfy the intrinsic hypotheses. □

Fix $M_T$ as reference and apply (2.1) to every member of $\Omega_T$. For $j \geq 0$, write $R_j = \mathbb{F}_q[X, Y, Z]_j$, and set $R_j = 0$ for $j < 0$. Put

$$\Delta_Q = 4\partial_X\partial_Z - \partial_Y^2, \qquad \mathcal{H}_d = \ker(\Delta_Q : R_d \longrightarrow R_{d-2}). \tag{3.3}$$

The operator $\Delta_Q$ is the Laplacian attached to the dual quadratic form. In the degrees used below, $\mathcal{H}_d$ is the top harmonic summand, while powers of $Q$ give radial summands.

**Lemma 3.7** (The edge-selected alternating cycle)**.** *For $q = 7, 11$, let $\Omega_+ \sqcup \Omega_-$ be the two one-factorization sheets. Every endpoint edge $e$ lies in a unique pair $M_+(e) \in \Omega_+$, $M_-(e) \in \Omega_-$. The matchings share only $e$, and after deleting it their union is one alternating cycle on the other $q - 1$ endpoints. The corresponding conic quotient has nonzero deepest radial projection.*

*Proof.* Move $e$ to $\{0, \infty\}$, so its secant is $Y$, and put

$$n = \frac{q-1}{2}, \qquad Q_0 = (\mathbb{F}_q^\times)^2.$$

The complementary matchings have the common torus normal form

$$\{\{u, cu\} : u \in Q_0\}, \qquad \{\{u, c^{-1}u\} : u \in Q_0\}, \tag{3.3a}$$

where $c^2 = 1/4$. Here is the promised derivation. The setwise edge stabilizer in $G^+$ is

$$D_e = \{x \mapsto sx : s \in Q_0\} \sqcup \{x \mapsto rs/x : s \in Q_0\},$$

where $r$ is any nonsquare. It has order $q-1$ and acts regularly on $\mathbb{F}_q^\times$: neither a nonidentity multiplication nor an inversion of the displayed form has a fixed point. Since each sheet is a one-factorization, its unique matching through $e$ is $D_e$-invariant.

One labeled calculation identifies the relevant involutions. For $q = 7$, $x \mapsto x/(2 - x)$ sends the edge $\{0, 2\}$ of $M_{B_3}$ to $e$ and the complement to

$$\{\{1, 5\}, \{2, 3\}, \{4, 6\}\} = \{\{u, 5u\} : u \in Q_0\}.$$

For $q = 11$, $x \mapsto x/(1 - x)$ sends $\{0, 1\}$ of $M_{H_3}$ to $e$ and the complement to

$$\{\{1, 2\}, \{3, 6\}, \{4, 8\}, \{5, 10\}, \{7, 9\}\} = \{\{u, 2u\} : u \in Q_0\}.$$

Choose $c = 3$ for $q = 7$ and $c = 6$ for $q = 11$. In each case $c$ is the nonsquare root of $c^2 = 1/4$, and the displayed complement is the $c^{-1}$-matching. Multiplication by any nonsquare fixes $e$, exchanges the two sheets, and rewrites that matching as the $c$-matching. This proves (3.3a), with $c$ specifically nonsquare. Multiplication by $c^{-2} = 4$ has order $n$ on $Q_0$ for $q = 7, 11$, proving the alternating-cycle assertion.

Write

$$A_c(X, Y, Z) = \prod_{u \in Q_0} (cu^2X - (1 + c)uY + Z).$$

The two matching products share the factor $Y$, and equality on the conic gives

$$A_{c^{-1}} - A_c = (XZ - Y^2)\Psi_c. \tag{3.3b}$$

If

$$h_j = \frac{n}{n-j}\binom{n-j}{j},$$

the resultant with $u^n - 1$ gives one Dickson recurrence:

$$A_{c^{-1}} - A_c = 2\sum_{j=0}^{(n-1)/2} (-1)^j h_j c^j (1+c)^{n-2j}(XZ)^j Y^{n-2j}. \tag{3.3c}$$

For $q = 7$, division in (3.3b) gives $\Psi_c = 5Y$. For $q = 11$, it gives

$$\Psi_c = 2Y^3 + XZY, \qquad \Delta_Q \Psi_c = 3Y.$$

The full quotient is $Y\Psi_c$, up to the fixed nonzero normalization. The common secant $Y$ has nonzero dual conic norm, so the displayed linear deepest traces force a nonzero radial Fischer component in both degrees. □

*Remark* 3.8 (The Paley carrier)*.* Label either sheet by its regular translation subgroup. The cross-sheet incidence matrix is the circulant support of $\{0\} \cup (\mathbb{F}_q^\times)^2$, the complement of the Paley difference set. Its bordered sign matrix is the Paley Hadamard matrix of order $q+1$. This explains the multiplier 4 and the sheet orientation, but it is not the Gorenstein pairing used below.

**Theorem 3.9** (The $3, 6, 10$ quotient ranks)**.** *For the matching configurations in (3.1), let*

$$W_T = \operatorname{span}_{\mathbb{F}_q}\{\Phi_{M_T}(M) : M \in \Omega_T\}.$$

*Then*

$$\begin{array}{cccc} T & \deg\Phi & W_T & \dim W_T \\ \hline A_3 & 1 & R_1 = \mathcal{H}_1 & 3, \\ B_3 & 2 & R_2 = \mathcal{H}_2 \oplus \mathbb{F}_7 Q & 6, \\ H_3 & 4 & \mathcal{H}_4 \oplus \mathbb{F}_{11} Q^2 & 10. \end{array} \tag{3.4}$$

*If $h_T = 4, 6, 10$ is the Coxeter number and $d = h_T/2 - 1$, the three rows have the uniform form*

$$W_T = \begin{cases} \mathcal{H}_d, & d \text{ odd}, \\ \mathcal{H}_d \oplus \mathbb{F}_q Q^{d/2}, & d \text{ even}, \end{cases} \qquad \dim W_T = h_T - 1 + \mathbf{1}_{\{d \text{ even}\}}.$$

*In particular, the first two configurations fill the complete quotient form space. The $H_3$ configuration spans a canonical ten-dimensional subspace of the fifteen-dimensional space $R_4$.*

*Proof mode.* The quotient construction and all three spanning assertions are proved symbolically. Irreducibility forces the $A_3$ image and the top $B_3$ summand. For $H_3$, a Sylow-normalizer cohomology argument excludes the middle Fischer layer and forces the top harmonic summand. The common edge-selected alternating cycle supplies both radial lines. A separate implementation reconstructs the three full quotient matrices and every edge-selected cycle from the data in (3.1).

*Proof.* A matching in (3.1) has $m = 3, 4, 6$ edges, respectively, so Proposition 2.1 places its quotient in $R_{m-2}$, of degrees $1, 2, 4$. Direct differentiation gives

$$\dim \mathcal{H}_1 = 3, \qquad \dim \mathcal{H}_2 = 5, \qquad \dim \mathcal{H}_4 = 9.$$

The radial vector $Q \in R_2$ is not harmonic in characteristic 7.

We first prove the $A_3$ and $B_3$ rows without quotient matrices. For $q = 5, 7$, put

$$G = \mathrm{PGL}_2(q), \qquad G^+ = \mathrm{PSL}_2(q), \qquad \chi(g) = \det(g)^{(q-1)/2}.$$

The Dickson form $s^q t - st^q$, which is the common conic restriction of the matching products, has contragredient weight $\det^{-1}$. The Clebsch identification and Fischer decomposition therefore give

$$\begin{array}{ll} q = 5 & \widetilde{R}_1 = R_1 \otimes \det^{-1} = M_2, \\ q = 7 & \widetilde{R}_2 = R_2 \otimes \det^{-1} = M_4 \oplus M_0, \end{array} \tag{3.4a}$$

where, uniformly,

$$M_r = \mathrm{Sym}^r(V^\vee) \otimes \overset{r/2}{\det} \otimes \chi, \qquad M_0 = \chi.$$

The scalar action is trivial, so these are $G$-modules. On $G^+$, $M_2$ and $M_4$ are the absolutely irreducible symmetric powers of highest weights $2 < 5$ and $4 < 7$, respectively [15, Fact 3.1].

For either type let

$$c(g) = \Phi_{M_T}(gM_T).$$

It is an affine cocycle, $c(gh) = gc(h) + c(g)$, and its value span is $G$-stable because

$$b\, c(g) = c(bg) - c(b). \tag{3.4b}$$

Distinct matchings have distinct products: unique factorization recovers the normalized secant factors. Hence every nonbase orbit point gives a nonzero cocycle value. In type $A_3$, irreducibility of $M_2$ now forces the value span to be all of $R_1$.

In type $B_3$, project the cocycle to $M_0 = \chi$. Its restriction to $G^+$ is a homomorphism from the simple group $\mathrm{PSL}_2(7)$ to the additive group of $\mathbb{F}_7$, and is therefore zero. A nonbase point on the $G^+$-sheet consequently gives a nonzero vector in $M_4$; irreducibility and (3.4b) force the whole five-space $M_4 = \mathcal{H}_2$.

Lemma 3.7 supplies an outer-sheet value with nonzero radial projection. Since $\Delta_Q$ kills $M_4 = \mathcal{H}_2$, its value has nonzero $M_0 = \mathbb{F}_7 Q$-projection. The span already contains $M_4$, so it contains $M_0$ as well. Hence $W_{B_3} = R_2$, proving the first two rows of (3.4).

We prove the $H_3$ row equivariantly. Put $G = \mathrm{PGL}_2(11)$, $G^+ = \mathrm{PSL}_2(11)$, and let $\chi(g) = \det(g)^5$ be the square-class character. On the conic, the common restriction of the matching products is the Dickson form $s^{11}t - st^{11}$, of contragredient weight $\det^{-1}$. Let $V = \mathbb{F}_{11}^2$, with binary forms carrying the contragredient $\mathrm{GL}_2(11)$-action. The equivariant Fischer decomposition obtained from the Clebsch identification $\mathrm{Sym}^2 V \simeq \mathbb{F}_{11}^3$, together with the Dickson weight in the quotient transformation (2.2), gives the linearized module

$$\widetilde{R}_4 = R_4 \otimes \overset{-1}{\det} = M_8 \oplus M_4 \oplus M_0. \tag{3.5}$$

Explicitly, $g \in \mathrm{GL}_2(11)$ acts on the first expression by

$$g \cdot f = \det(g)^{-1}(f \circ \rho(g)^{-1}).$$

The three summands are

$$
\begin{aligned}
M_8 &= \mathcal{H}_4 \otimes \overset{-1}{\det} \simeq \mathrm{Sym}^8(V^\vee) \otimes \overset{-1}{\det}, \\
M_4 &= Q\mathcal{H}_2 \otimes \overset{-1}{\det} \simeq \mathrm{Sym}^4(V^\vee) \otimes \overset{-3}{\det}, \\
M_0 &= \mathbb{F}_{11}Q^2 \otimes \overset{-1}{\det} \simeq \overset{-5}{\det} = \chi.
\end{aligned} \tag{3.6}
$$

Equivalently, all three summands have the uniform form

$$
M_r = \mathrm{Sym}^r(V^\vee) \otimes \overset{r/2}{\det} \otimes \chi, \qquad r = 8, 4, 0.
$$

A scalar $\lambda I$ acts here as $\lambda^{-r}\lambda^r(\lambda^2)^5 = 1$, so these modules and their direct sum descend from $\mathrm{GL}_2(11)$ to $G$. This is the $G$-action used below. The determinant twist changes only the linearization, not the underlying Fischer subspaces $\mathcal{H}_4, Q\mathcal{H}_2, \mathbb{F}_{11}Q^2$ of $R_4$.

We need two elementary cohomology facts. Let $U = \langle u \rangle$ be the upper-unipotent Sylow 11-subgroup and let $a$ generate the diagonal torus of order 10. Since $r < 10$, the cyclic norm $1 + u + \cdots + u^{10} = (u-1)^{10}$ vanishes on $M_r$, while $(u-1)M_r$ has codimension one. Thus $H^1(U, M_r)$ is represented by the class of $x^r$. Diagonal conjugation acts on that class by

$$
a^{4-r/2}. \tag{3.7}
$$

Restriction from $G$ to $U$ is injective because $[G : U] = 120$ is invertible in $\mathbb{F}_{11}$, and its image is fixed by the Sylow normalizer. Equation (3.7) therefore gives

$$
H^1(G, M_4) = H^1(G, M_0) = 0, \qquad \dim H^1(G, M_8) \le 1. \tag{3.8}
$$

The stabilizer $H = A_5$ of $M_{H_3}$ lies in $G^+$. The restrictions of $M_8, M_4, M_0$ to $H$ are respectively $4 \oplus 5, 5, 1$, so

$$
M_8^H = M_4^H = 0, \qquad \dim M_0^H = 1. \tag{3.9}
$$

Equivalently, average the characters $(9,1,0,-1,-1)$, $(5,1,-1,0,0)$, and $(1,1,1,1,1)$ over the $A_5$ class sizes $(1,15,20,12,12)$.

Let $c(g) = \Phi_{M_{H_3}}(gM_{H_3})$. The common-restriction identity gives

$$
c(gh) = gc(h) + c(g),
$$

and $c$ vanishes on $H$. Project $c$ through (3.5). By (3.8), its $M_4$-component is a coboundary; its translating vector lies in $M_4^H = 0$, so that component vanishes. This proves, without quotient coordinates,

$$
\Delta_Q \Phi_{M_{H_3}}(M) \in \mathbb{F}_{11}Q \quad \text{for every } M \in \Omega_{H_3}. \tag{3.10}
$$

Likewise the $M_0$-component is a coboundary and hence vanishes on $G^+$. Choose any other matching in the eleven-element $G^+$-orbit of $M_{H_3}$. Its quotient is nonzero: equality of the two secant products would identify their irreducible line factors and hence the matchings. Its $M_4$- and $M_0$-components vanish, so it gives a nonzero vector in $M_8$. The degree-eight restricted $\mathrm{SL}_2(\mathbb{F}_{11})$-module is irreducible [15, Fact 3.1]. The span of the cocycle values is $G$-stable, since

$$
b\,c(g) = c(bg) - c(b) \qquad (b, g \in G).
$$

Therefore the quotient span contains all of $M_8$.

Lemma 3.7 supplies an outer-sheet value with nonzero deepest radial projection. Since $\Delta_Q^2$ kills $M_8 \oplus M_4$, this supplies the $M_0$-projection. The span already contains $M_8$, so it also contains $M_0$. Hence $W_{H_3} = \mathcal{H}_4 \oplus \mathbb{F}_{11}Q^2$, of dimension 10. Since $\dim R_4 = 15$, this equality is proper. □

*Remark* 3.10 (Historical radial coordinates). Direct coordinates give $\Delta_Q\Phi = 4$ for the displayed $B_3$ pair and $\Delta_Q^2\Phi = 10$ for the displayed $H_3$ pair. These values independently check the two specializations of Lemma 3.7; they are not premises of its Dickson-recurrence proof.

**Corollary 3.11** (The base-choice class and its homogeneous extension). *Let $c_8 : G \to M_8$ be the top-harmonic projection of the base-choice cocycle $c$. Then*

$$H^1(G, M_8) = \mathbb{F}_{11}[c_8]. \tag{3.11a}$$

*In particular, the affine $H_3$ quotient configuration has no $G$-equivariant origin: no translation makes its affine action linear. Equivalently, the homogeneous module*

$$E_{c_8} = M_8 \oplus \mathbb{F}_{11}, \qquad g \cdot (v, t) = (gv + t\, c_8(g), t), \tag{3.11b}$$

*is nonsplit. Up to equivalence and nonzero rescaling of the quotient coordinate, it is the unique nonsplit extension of the trivial $G$-module by $M_8$.*

*Proof.* Equation (3.8) gives $\dim H^1(G, M_8) \leq 1$. On the base sheet the $M_0$-component vanishes, and the $M_4$-component vanishes by (3.9); the nonzero same-sheet quotient used in the proof above is therefore a value of $c_8$. If $c_8(g) = gv - v$, its vanishing on $H = \mathrm{Stab}_G(M_{H_3})$ would put $v$ in $M_8^H = 0$, forcing $c_8 = 0$, a contradiction. Thus $[c_8] \neq 0$, proving (3.11a).

Changing the reference quotient translates the affine configuration and changes $c_8$ by a coboundary. Hence a $G$-equivariant origin would kill $[c_8]$, which is impossible. Finally the standard identification $\mathrm{Ext}^1_{\mathbb{F}_{11}G}(\mathbb{F}_{11}, M_8) \simeq H^1(G, M_8)$ identifies (3.11b) with $[c_8]$; the one-dimensionality just proved gives the last assertion. □

**Corollary 3.12** (The exact $H_3$ loss). *Over $\mathbb{F}_{11}$, the quartic form space has the Fischer decomposition*

$$R_4 = \mathcal{H}_4 \oplus Q\mathcal{H}_2 \oplus \mathbb{F}_{11}Q^2. \tag{3.12}$$

*Consequently,*

$$R_4 = W_{H_3} \oplus Q\mathcal{H}_2, \qquad R_4/W_{H_3} \simeq Q\mathcal{H}_2. \tag{3.13}$$

*Equivalently,*

$$W_{H_3} = \{f \in R_4 : \Delta_Q f \in \mathbb{F}_{11}Q\}. \tag{3.14}$$

*Thus the $H_3$ quotient configuration retains the top harmonic nine-space and the deepest radial line, and omits precisely the five-dimensional middle radial layer.*

*Proof.* For a homogeneous form $f$ of degree $r$, direct differentiation gives

$$\Delta_Q(Qf) = (4r + 6)f + Q\Delta_Q f. \tag{3.15}$$

Hence $\Delta_Q(Qh) = 3h$ for $h \in \mathcal{H}_2$, while $\Delta_Q(Q^2) = 9Q$ over $\mathbb{F}_{11}$. Applying $\Delta_Q^2$ and then $\Delta_Q$ shows that the three summands in (3.12) meet trivially. Their dimensions are $9, 5, 1$, so they exhaust the fifteen-dimensional space $R_4$. Theorem 3.9 then gives (3.13). If $f = h_4 + Qh_2 + cQ^2$ is its decomposition, then

$$\Delta_Q f = 3h_2 + 9cQ.$$

The quadratic decomposition $R_2 = \mathcal{H}_2 \oplus \mathbb{F}_{11}Q$ therefore gives (3.14). □

The rank proof has one general part and one exceptional input. The conic ideal gives the common restriction and quotient, while irreducibility forces the $A_3$ image and the top $B_3$ summand. For $H_3$, (3.7)–(3.10) use the specific $A_5 < \mathrm{PGL}_2(11)$ module data to force the top summand and exclude the middle one. Lemma 3.7 supplies both radial lines from the same recurrence. Appendix A records the detector calculations and their finite seams.

Corollary 3.12 identifies the linear information lost by the $H_3$ orbit, but linear span does not recover the two factorization sheets inside the finite configuration. Their recovery begins one categorical level higher, with products of affine evaluation functions.

# 4 Balanced sheets and cubic orientation

For $T = B_3, H_3$, put $q = 7, 11$, respectively. The subgroup

$$G^+ = \mathrm{PSL}_2(q) \subset G = \mathrm{PGL}_2(q)$$

splits $\Omega_T$ into two orbits $\Omega_+ \sqcup \Omega_-$, each of size $q$; an element of the outer coset exchanges them. These are the two one-factorization sheets. The $A_3$ orbit does not split: its $S_4$ stabilizer cannot lie in $\mathrm{PSL}_2(5) \simeq A_5$, already by order.

Write $x_M = \Phi_{M_T}(M) \in W_T$. Let $L \subseteq \mathbb{F}_q^{\Omega_T}$ be the evaluation space of affine-linear functions on the finite configuration $\{x_M\}$. Since the points affinely span $W_T$,

$$\dim L = 1 + \dim W_T = q.$$

For subspaces of functions, write

$$L^{\circ 2} = \mathrm{span}\{fg : f, g \in L\},$$

where multiplication is pointwise. The common second-moment form in the proposition below is

$$B(\lambda, \mu) = \sum_{M \in \Omega_+} \lambda(x_M)\mu(x_M) = \sum_{M \in \Omega_-} \lambda(x_M)\mu(x_M)$$

on $W_T^\vee$. A *field-valued strength-two trade* is a weight $\tau \in k^\Omega$ satisfying, for all $f, g \in L$,

$$\sum_{M \in \Omega} \tau(M) f(M) g(M) = 0.$$

**The $B_3$ model.** Here the quotient configuration consists of fourteen points spanning the six-space $W_{B_3}$, split into two seven-point sheets. Its affine evaluation space $L$ has dimension seven. Restriction to either sheet has rank six, while the common second-moment form has rank five and a radical functional that takes distinct constant values on the two sheets. It follows that $L^{\circ 2}$ is the thirteen-dimensional hyperplane of functions with equal sheet sums; its one-dimensional annihilator is the sheet sign. Thus the quadratic algebra recovers the 7+7 partition without a search through partitions. The first nonzero signed moment is cubic, so $L^{\circ 3} = \mathbb{F}_7^{14}$, and the homogenized points have $h$-vector $(1, 6, 6, 1)$. The next proposition isolates the quadratic step in a form that applies to both balanced configurations.

**Proposition 4.1** (Radical–Hadamard recovery)**.** *Let $k$ be a field and $\Omega = \Omega_+ \sqcup \Omega_-$, with $|\Omega_+| = |\Omega_-| = q$ and $q \cdot 1_k = 0$. Suppose an affine evaluation space $L \subseteq k^\Omega$ has the following properties:*

(i) *each sheet has zero first moment, and the two sheets have equal second moments;*

(ii) $\dim L = q$, *and restriction from* $L$ *onto the zero-sum hyperplane of either sheet has rank* $q-1$*;*

(iii) *the second-moment form on linear functionals has rank* $q-2$ *and a one-dimensional radical;*

(iv) *a nonzero radical functional is constant on each sheet, with different constants on the two sheets.*

*Then*

$$L^{\circ 2} = \left\{ (u_+, u_-) : \sum_{\Omega_+} u_+ = \sum_{\Omega_-} u_- \right\}, \tag{4.1}$$

*and*

$$(L^{\circ 2})^{\perp} = k(\mathbf{1}_{\Omega_+} - \mathbf{1}_{\Omega_-}). \tag{4.2}$$

*Consequently every field-valued strength-two trade is a scalar multiple of the sheet sign.*

*Proof.* Let $e_+$ and $e_-$ be the two sheet indicators. The evaluation vector of a nonzero radical functional is $a_+e_+ + a_-e_-$, with $a_+ \neq a_-$. Together with $1 = e_+ + e_- \in L$, it therefore yields $e_+, e_- \in L$.

Let $H_\pm$ be the zero-sum hyperplane on $\Omega_\pm$. The first-moment hypothesis and $q \cdot 1_k = 0$ put each restriction of $L$ inside $H_\pm$; the rank hypothesis makes both restrictions surjective. Multiplication by $e_+$ and $e_-$ now gives

$$(H_+ \oplus 0) + (0 \oplus H_-) \subseteq L^{\circ 2}. \tag{4.3}$$

This subspace has dimension $2q-2$.

For $f, g \in L$, the difference between the two sheet sums of $fg$ expands into constant, first-moment, and second-moment terms. The hypotheses make all three differences zero, so $L^{\circ 2}$ lies in the hyperplane on the right of (4.1). Finally choose restrictions $a, b \in H_+$ with nonzero standard pairing and lift them to $f, g \in L$. Equality of the second moments gives the same nonzero pairing on $\Omega_-$. Thus $fg$ has equal nonzero sheet sums and does not lie in (4.3). We have found $2q-1$ independent directions in the $2q-1$-dimensional hyperplane, proving (4.1). Its annihilator under the standard coordinate pairing is the line in (4.2). $\square$

**Proposition 4.2** (The modular-permutation mechanism)**.** *For* $T = B_3, H_3$*, the affine quotient configuration satisfies all four hypotheses of Proposition 4.1. More precisely, write*

$$W_T = U_T \oplus \mathbb{F}_q r, \qquad U_{B_3} = \mathcal{H}_2, \quad U_{H_3} = \mathcal{H}_4.$$

*On either q-point sheet, affine evaluation has image the zero-sum hyperplane and rank* $q-1$*. The common second-moment form has rank* $q-2$*, with radical the radial functional* $r^\vee$*; that functional is constant on each sheet and its two constants are distinct.*

*Proof.* As a $G^+$-module, $U_T$ is the irreducible module $M_{q-3}$ of dimension $q-2$. Project the affine quotient points to $U_T$, writing the resulting sheet as $\{y_\omega\}$. The cocycle argument in Section 3 says that the differences $y_\omega - y_{\omega'}$ span $U_T$. Moreover $\sum_\omega y_\omega$ is $G^+$-fixed: translating the sheet adds $q$ times the cocycle value, which vanishes in characteristic $q$. Since $U_T^{G^+} = 0$,

$$\sum_\omega y_\omega = 0.$$

Let $P = \mathbb{F}_q^{\Omega_+}$, and put

$$P_0 = \left\{ (a_\omega) : \sum_\omega a_\omega = 0 \right\}, \qquad S = \mathbb{F}_q \mathbf{1}.$$

Here $S \subset P_0$, the defining-characteristic feature that drives the argument. The map

$$P_0 \longrightarrow U_T, \qquad (a_\omega) \longmapsto \sum_\omega a_\omega y_\omega$$

is onto because the differences span, and the preceding moment identity kills $S$. Both $P_0/S$ and $U_T$ have dimension $q-2$, so it induces an isomorphism

$$P_0/S \simeq U_T.$$

The standard dot product on $P_0$ has radical exactly $S$. Dualizing therefore shows that the evaluations of $U_T^\vee$ form a nondegenerate $(q-2)$-space complementary to $S$ in $P_0$. Adding constants gives all of $P_0$, proving the restriction rank and the top second-moment rank. The same argument applies to the other sheet.

The two top second moments are equal. On

$$M_{q-3} = \mathrm{Sym}^{q-3}(V^\vee) \otimes \overset{(q-3)/2}{\det} \otimes \chi$$

the standard apolar pairing is $G$-invariant: its determinant weight $\det^{-(q-3)}$ is cancelled by the square of the displayed twist, and $\chi^2 = 1$. Each sheet moment is a nonzero $G^+$-invariant form, hence a scalar multiple of this pairing, while an outer element transports one sheet to the other by an apolar isometry. Their scalars, and thus the two forms, are equal.

Finally, projection of the affine cocycle to the radial character $\chi$ restricts on the simple group $G^+$ to an additive homomorphism, hence vanishes. The radial coordinate is consequently zero on the base sheet and constant on the other. The secant-norm calculation in Lemma 3.7 shows that the outer constant is nonzero in both types. Because $q \cdot 1_{\mathbb{F}_q} = 0$, radial–radial and radial–top second moments vanish. Thus $r^\vee$ is exactly the one-dimensional radical, and its sheet constants are distinct. □

Let $T = B_3, H_3$, with $q = 7, 11$, respectively, and fix a reference matching $M_T$ with stabilizer $K = S_4, A_5$. Put $m = (q+1)/2$ and $d = m-2$. Write $\mathcal{A}_T$ for the affine space of degree-$m$ forms whose restriction to the conic is the fixed binary form vanishing on all $q+1$ marked rational points. We call its members that split completely into linear factors over $\overline{\mathbb{F}}_q$ the *geometric Chow locus.* All intersections below are set-theoretic. Symmetry leaves one affine parameter in $\mathcal{A}_T$; geometric splitting selects one rational point.

**Theorem 4.3** (The fixed line and its Chow point)**.** *With the preceding notation and the determinant normalization of Section 2, the following hold.*

(i) *The $K$-fixed locus is an affine line:*

$$\mathcal{A}_T^K = P_{M_T} + Q(R_d)^K = P_{M_T} + \mathbb{F}_q Q^{d/2+1}. \tag{4.4a}$$

*Its $q$ rational points lie in distinct $G$-orbits, each with stabilizer $K$.*

(ii) *The rational fixed line meets the geometric Chow locus in the single point $P_{M_T}$. Equivalently, the matching point is its unique geometrically split point.*

(iii) *The orbit of each noncoalescent point has $2q$ points, and its quadratic-trade space is the sheet-sign line. Consequently $q-2$ of these orbits satisfy the same one-dimensional two-valued strength-two trade condition as the matching orbit but do not come from matchings.*

*Thus the quadratic-trade condition alone does not classify the $B_3/H_3$ configurations; the matching carrier in Theorem 3.6 is essential. Within that carrier, the classification remains exact.*

*Proof.* The translation space of $\mathcal{A}_T$ is $QR_d$, and $P_{M_T}$ is $K$-fixed. Since $q \nmid |K|$, invariant dimensions may be computed by averaging characters. For $B_3$, the invariant subspace of the quadratic form module under the irreducible orthogonal $S_4$-action is $\mathbb{F}_7 Q$. For $H_3$, the character of the quartic form module on the classes of orders $1, 2, 3, 5, 5$ has values $15, 3, 0, 0, 0$; hence

$$\dim R_4^{A_5} = \frac{15 + 15 \cdot 3}{60} = 1,$$

and $R_4^{A_5} = \mathbb{F}_{11} Q^2$. This proves (4.4a).

The proof of Theorem 3.6 shows that $N_G(K) = K$. A point on (4.4a) cannot be fixed by $G^+$: its difference from $P_{M_T}$ is radial and hence $G^+$-fixed, which would make $P_{M_T}$ itself $G^+$-fixed. Maximality of $K$ in $G^+$, together with $N_G(K) = K$, therefore makes its stabilizer exactly $K$. If two points of the line were $G$-conjugate, a conjugating element would normalize $K$, hence lie in $K$, and the points would be equal. This proves the last assertion of (i).

Suppose a $K$-fixed point of $\mathcal{A}_T$ is a product of linear forms. Each factor cuts a degree-two divisor on the conic, while their product cuts the squarefree divisor of all $q + 1$ marked points. Thus the factors pair those points into marked secants. This remains true when the factorization is taken over $\overline{\mathbb{F}}_q$: every factor divisor is a two-element subset of the displayed rational divisor, so its joining line is rational. Unique factorization makes that matching $K$-invariant. For $S_4$ on eight points, a point stabilizer is $C_3$ and the unique possible two-point block stabilizer is $N_{S_4}(C_3) = S_3$. For $A_5$ on twelve points, the corresponding groups are $C_5$ and $N_{A_5}(C_5) = D_{10}$. Thus in either case the $K$-invariant matching is unique, proving (ii).

It remains to track the quadratic relation, not the individual orbits. Write the fixed line in quotient coordinates as $x_t = x_0 + tr$, where $r = Q^{d/2}$ spans the radial character. On $G^+$ this character is trivial, while the outer coset negates it. Relative to the moving base point $x_t$, the top harmonic configurations on both sheets are therefore unchanged; one radial sheet constant remains 0, and the other changes from the nonzero constant $c$ of Proposition 4.2 to $c - 2t$. Because $q \cdot 1_{\mathbb{F}_q} = 0$ and each top sheet has zero sum, this translation changes neither the first moments nor the two equal second moments.

There is exactly one coalescence parameter $t = c/2$. For every other $t$, all four hypotheses of Proposition 4.1 are unchanged, so the quadratic-trade space is exactly the sheet-sign line. The matching point is $t = 0$, and $c \neq 0$, so it is not the coalescence point. Among the $q - 1$ noncoalescent orbits, exactly one is the matching orbit by (ii); the remaining $q - 2$ prove (iii). $\square$

*Remark* 4.4 (The coalescence point). At $t = c/2$, the two radial constants agree and hypothesis (iv) of Proposition 4.1 fails. The square rank at this single point is not needed for the classification or for Theorem 4.3.

For reference, exact reconstruction of the two configurations gives

| $T$ | $q$ | sheet restriction ranks | second-moment rank | $\dim L^{\circ 2}$ |
|---|---|---|---|---|
| $B_3$ | 7 | 6, 6 | 5 | 13, |
| $H_3$ | 11 | 10, 10 | 9 | 21. |

(4.4)

In each row the radical is one-dimensional and takes two distinct sheet values. Proposition 4.2, rather than this table, proves these facts. The exact certificate and its independent reconstruction are cross-checks only; no enumeration of equal-size subsets or row reduction is used in the proof.

**Lemma 4.5** (A full-support quadratic relation forces the cube)**.** *Let $\Omega$ be finite and let $L \subseteq k^{\Omega}$ be a unital linear space with* $\dim L > 1$*. Suppose*

$$L^{\circ 2} = \left\{ u : \sum_{\omega \in \Omega} \epsilon_{\omega} u_{\omega} = 0 \right\}$$

*for some $\epsilon \in (k^{\times})^{\Omega}$. Then*

$$L^{\circ 3} = k^{\Omega}.$$

*Proof.* If a nonzero vector $\eta$ annihilated $L^{\circ 3}$, then $L^{\circ 2} \subseteq L^{\circ 3}$ would put $\eta$ in $k\epsilon$. For every $f \in L$, the vector $\eta \circ f$ also annihilates $L^{\circ 2}$, because

$$\langle \eta \circ f, h \rangle = \langle \eta, fh \rangle = 0 \qquad (h \in L^{\circ 2}).$$

Thus $\eta \circ f \in k\epsilon = k\eta$. Since $\eta$ has full support, every $f \in L$ is constant, contrary to $\dim L > 1$. The annihilator of $L^{\circ 3}$ is therefore zero. ☐

**Theorem 4.6** (Balanced sheets and cubic-first orientation)**.** *For $T = B_3, H_3$, the unordered pair $\{\Omega_+, \Omega_-\}$ is intrinsic to the affine quotient configuration:*

(i) *up to interchange, $\Omega_+$ and $\Omega_-$ are the only complementary halves with equal first and second tensor moments;*

(ii) *if $\epsilon$ is $+1$ on $\Omega_+$ and $-1$ on $\Omega_-$, then*

$$\mu_j = \sum_{M \in \Omega_T} \epsilon(M) x_M^{\odot j} \in \mathrm{Sym}^j(W_T)$$

*satisfies*

$$\mu_1 = 0, \qquad \mu_2 = 0, \qquad \mu_3 \neq 0; \tag{4.5}$$

(iii) *$\mu_3$ is independent of the reference matching, $G^+$ fixes it, and the outer coset negates it. Within $G$,*

$$\mathrm{Stab}_G(\mu_3) = G^+, \qquad \mathrm{Stab}_G(\mathbb{F}_q \mu_3) = G. \tag{4.6}$$

*Thus degree three is the first signed tensor moment, or linear functional of such a moment, that orients the recovered sheet pair. This minimality statement concerns signed tensor moments, not every possible nonlinear statistic of the configuration.*

*Proof mode.* Proposition 4.1 is a symbolic proof. Proposition 4.2 proves its hypotheses from the defining-characteristic permutation module and the shared alternating-cycle witness. Lemma 4.5 forces the nonzero cubic without a tensor calculation or Gorenstein premise. Signed Gale duality records self-association, while maximal-isotropic quotient duality gives the Gorenstein pairing. Exact finite arithmetic independently cross-checks the displayed ranks.

*Proof.* Equality of first and second tensor moments for a complementary pair is equivalent, by polarization, to its sign weight annihilating $L^{\circ 2}$. Since the characteristics are 7 and 11, polarization through degree two is valid. Proposition 4.1 says that the annihilator is exactly the sheet-sign line. A sign weight taking only the values $\{\pm 1\}$ is therefore $\epsilon$ or $-\epsilon$, proving (i).

The same moment equalities give $\mu_1 = \mu_2 = 0$. Proposition 4.1 identifies $L^{\circ 2} = \ker\langle \epsilon, -\rangle$, and $\epsilon$ has full support. Lemma 4.5 gives $L^{\circ 3} = \mathbb{F}_q^{\Omega_T}$. Hence $\epsilon$ cannot annihilate all cubic products. Since the lower signed moments vanish and 6 is invertible in $\mathbb{F}_7, \mathbb{F}_{11}$, polarization gives $\mu_3 \neq 0$.

For the geometric consequence, let $A$ be the $q$-by-$2q$ matrix whose $M$-th column is $\widehat{x}_M = (1, x_M)$, and put $D = \operatorname{diag}(\epsilon(M))$. Equal sheet sizes and the vanishing signed moments through degree two give

$$ADA^{\mathsf{T}} = 0.$$

The affine spanning statement gives $\operatorname{rank} A = q$, so the rows of $AD$ form the kernel of $A$. Thus $AD$ is a Gale matrix, and its columns differ from those of $A$ only by the nonzero scalars $\epsilon(M)$. The homogenized configuration is self-associated.

We now prove the Gorenstein statement directly. On

$$V = \mathbb{F}_q^{\Omega_+} \oplus \mathbb{F}_q^{\Omega_-}$$

use the nondegenerate signed coordinate form

$$B(u, v) = \sum_{\Omega_+} u_M v_M - \sum_{\Omega_-} u_M v_M. \tag{4.6a}$$

Proposition 4.1 identifies

$$L^{\circ 2} = \langle \mathbf{1} \rangle^{\perp_B}. \tag{4.6b}$$

For $u, v \in L$, the product $u \circ v$ lies in the right side of (4.6b), so $B(u, v) = 0$. Thus $L$ is isotropic. Since $\dim L = q = \frac{1}{2} \dim V$, it is maximal isotropic:

$$L = L^{\perp_B}. \tag{4.6c}$$

Let $\overline{R}$ be the reduction of the homogeneous coordinate ring by the homogenizing coordinate. Its positive graded pieces are

$$\overline{R}_1 = L/\langle \mathbf{1} \rangle, \qquad \overline{R}_2 = L^{\circ 2}/L, \qquad \overline{R}_3 = V/L^{\circ 2}.$$

The form $B$ descends to

$$\overline{R}_1 \times \overline{R}_2 \longrightarrow \mathbb{F}_q. \tag{4.6d}$$

Its left radical is $(L^{\circ 2})^{\perp}_B/\langle \mathbf{1} \rangle = 0$, and its right radical is $L^{\perp}_B/L = 0$. Hence (4.6d) is perfect. Lemma 4.5 gives $L^{\circ 3} = V$, so $\overline{R}_3$ is one-dimensional and multiplication realizes (4.6d) as the degree-1-by-degree-2 socle pairing. Therefore $\overline{R}$ is Artinian Gorenstein with Hilbert function

$$(1, q-1, q-1, 1).$$

The homogenizing coordinate is regular because it is nonzero at every point, so the projective coordinate ring is arithmetically Gorenstein. The unique quadratic dependence is $\epsilon$, which has full support; hence every one-point deletion imposes independent conditions on quadrics.

If the reference matching changes, every point translates by one vector $c$. Expanding the signed third moment after $x_M \mapsto x_M + c$ leaves $\mu_3$ unchanged: the other terms contain $\mu_2, \mu_1$, or $\mu_0 = \sum_M \epsilon(M) = 0$.

Every element of $G^+$ preserves both sheets and hence fixes the signed sum. Every element of the outer coset exchanges the sheets and negates it. Since $\mu_3 \neq 0$, these two transformation laws give (4.6). Finally (4.5) proves the stated minimality inside the signed tensor-moment family. $\square$

**Corollary 4.7** (Graded evaluation algebra)**.** *Put $L^{\circ 0} = \langle 1 \rangle$, and let $L^{\circ d}$ be the span of the pointwise products of $d$ elements of $L$. For $T = B_3, H_3$, the pointwise powers of the affine evaluation space satisfy*

$$\dim L^{\circ d} = \begin{cases} 1, & d = 0, \\ q, & d = 1, \\ 2q-1, & d = 2, \\ 2q, & d \geq 3. \end{cases}$$

*In particular, $L^{\circ 3} = \mathbb{F}_q^{\Omega_T}$. The homogenized configuration*

$$\widehat{X}_T = \{[1 : x_M] : M \in \Omega_T\} \subseteq \mathbb{P}^{q-1}$$

*therefore has Hilbert function*

$$1,\ q,\ 2q-1,\ 2q,\ 2q, \ldots$$

*and $h$-vector $(1, q-1, q-1, 1)$.*

*Proof.* Pair functions on $\Omega_T$ with the sheet sign by $\langle \epsilon, f \rangle = \sum_M \epsilon(M) f(M)$. Proposition 4.1 gives $L^{\circ 2} = \ker\langle \epsilon, - \rangle$, with $\epsilon$ of full support. Lemma 4.5 gives $L^{\circ 3} = \mathbb{F}_q^{\Omega_T}$. Multiplication by 1 gives the same equality in every higher degree. Homogeneous degree-$d$ forms on $\widehat{X}_T$ restrict to $L^{\circ d}$, which gives the Hilbert function; its first difference is the stated $h$-vector. Changing the reference matching translates every $x_M$. Such a translation preserves $L$ and all its pointwise powers and acts projectively on $\widehat{X}_T$, so the graded algebra and Hilbert function are independent of the reference. $\square$

**Corollary 4.8** (Self-association and cubic duality)**.** *For $T = B_3, H_3$, the configuration $\widehat{X}_T \subseteq \mathbb{P}^{q-1}$ is a reduced self-associated arithmetically Gorenstein set of $2q$ points. Every one-point deletion imposes independent conditions on quadrics, so $\widehat{X}_T$ has the Cayley–Bacharach property in degree two. Its socle residue vector is the sheet sign $\epsilon$. The Artinian reduction by the homogenizing coordinate has Hilbert function $(1, q-1, q-1, 1)$, and its Macaulay inverse system is the cubic line $\mathbb{F}_q \mu_3$.*

*Proof.* The proof of Theorem 4.6 already establishes signed Gale self-duality, the Cayley–Bacharach property in degree two, the Gorenstein coordinate ring, and the displayed $h$-vector. These arguments commute with base change to an algebraic closure, so the Gorenstein conclusion descends to $\mathbb{F}_q$.

Let $z$ be the homogenizing coordinate. It is nonzero at every point and is therefore regular on the coordinate ring. The degree $d$ part of the reduction modulo $z$ is $L^{\circ d}/L^{\circ(d-1)}$, so its Hilbert function is the $h$-vector from Corollary 4.7. The functional

$$f \longmapsto \sum_M \epsilon(M) f(x_M)$$

annihilates degrees below three and is nonzero in degree three. Thus $\epsilon$ is the socle residue vector. Degree-three polarization identifies its Macaulay dual generator with $\mu_3$. $\square$

After base change to an algebraic closure, the signed coordinate form can be put in the standard form used for self-dual codes. In that form, the implication above is the zero-defect case of the modern self-dual-code criterion: the Gorenstein defect is measured by the defect of the Schur square, or equivalently by decomposability [13, Theorem 3.11 and Corollaries 3.12–3.13]. Here $\dim L^{\circ 2} = 2q-1$ already places both configurations in the zero-defect case. The ideal, deletion, socle, inverse-system, and minimal-resolution calculations are also checked directly in the verification bundle. Self-association does not select an affine origin. Translating all $x_M$ acts projectively on the columns of $A$ and preserves the signed Gale identity, in agreement with the nontrivial base-choice obstruction behind the quotient construction.

The cubic remembers an orientation that the linear quotient does not. Indeed, the signed linear relation

$$\sum_{M\in\Omega_+} x_M = \sum_{M\in\Omega_-} x_M$$

already shows that the sheet sign is killed by the linear quotient map. Quadratic products recover the unordered partition through (4.2), and the cubic supplies the sign exchanged when the two recovered levels are swapped.

**Corollary 4.9** (Secant-product tensor syzygies). *For $T = B_3, H_3$, let $m_Q : R_{m-2} \to R_m$ denote multiplication by $Q$. Regarding the displayed powers as symmetric tensors, not polynomial powers, the canonically scaled secant products satisfy*

$$\begin{aligned} \sum_M \epsilon(M) P_M &= 0, \\ \sum_M \epsilon(M) P_M^{\odot 2} &= 0, \\ \sum_M \epsilon(M) P_M^{\odot 3} &= \mathrm{Sym}^3(m_Q)(\mu_3) \neq 0. \end{aligned} \tag{4.7}$$

*Proof.* Substitute $P_M = P_{M_T} + Qx_M$ and expand. Each sheet has $q \cdot 1_{\mathbb{F}_q} = 0$, and the signed quotient moments vanish through degree two. All terms below cubic degree cancel, leaving $\mathrm{Sym}^3(m_Q)(\mu_3)$ in degree three. Since $\mathbb{F}_q[X, Y, Z]$ is a domain, multiplication by $Q$ is injective; over a field, the symmetric powers of an injective linear map are injective. □

# 5 Conclusion

Restriction to a conic erases the pairing of its marked points. Dividing differences of secant products by the conic equation retains enough information to recover exactly two balanced matching orbits: $B_3/\mathbb{F}_7$ and $H_3/\mathbb{F}_{11}$. The proof derives their one-factorization sheets from the two-valued quadratic trade and excludes all other odd prime powers without a matching census. The same quotient construction gives the ranks $3, 6, 10$; for $H_3$, it retains the top harmonic part and deepest radial line and loses exactly $Q\mathcal{H}_2$.

The matching hypothesis is the precise boundary. Each surviving stabilizer fixes an affine line of conic products. Its only completely reducible point is the matching product, although $q-2$ other orbits retain the same two-valued trade. The trade therefore remembers the homogeneous sheet module but not its embedding as a secant factorization; the Chow condition restores that information.

On the matching orbit, quadratic products recover the two sheets without orienting them. Their first signed tensor moment is cubic, and this cubic orients the recovered pair. It is also the Macaulay inverse system of the self-associated Gorenstein homogenization. The general Schur-square criterion explains Gorensteinness; the configuration-specific result is the chain from quadratic trade to matching orbit, from matching orbit to unordered sheets, and from unordered sheets to the cubic orientation. In these configurations the degree filtration therefore separates the information carried by the construction: quadratic data recognize and recover the unordered factorization, while the first odd degree both restores orientation and supplies the Macaulay dual generator.

## Clebsch: Rigidity from Sparse Shadows

I From deep holes, a form arises;
II **beneath it, paired chords turn true;**
III through paired veils, the two sheets exchange;
IV from bare, whispered words, a plane rises;
V beneath one form, two shadows, a hidden twist between.

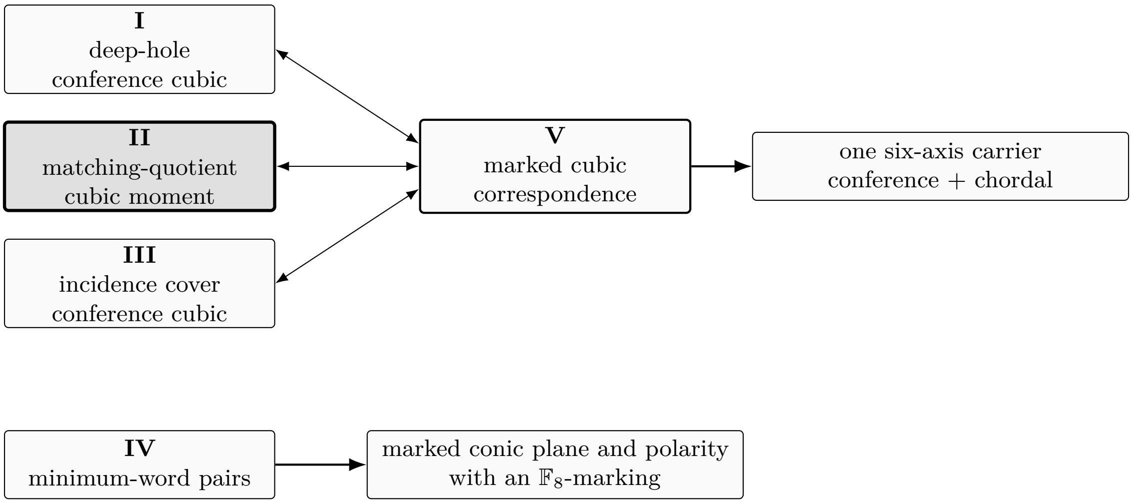


Figure 1: Programme map. The upper double arrows are Paper V's marked transports and returns between the chordal and conference shadows; the right arrow is its common-carrier reconstruction. The lower solid arrow is Paper IV's independent minimum-word reconstruction. Only the residual $C_2$- and Frobenius-orbit markings are compared across the two branches.

The five papers are logically independent; the map records reconstruction correspondences and thematic relations, not proof dependencies.

This paper enters the programme through the cubic signed moment produced by its matching quotient. Paper V later shows that, after the marked five-isotypic bridge to the six-axis carrier, the chosen-sheet projection of that moment is a metric chordal shadow. That placement is downstream of the classification proved here: the chordal geometry is not an input to the matching-quotient reconstruction.

**Appendix guide.** Appendix A is the load-bearing specialist layer for the uniform modular exclusion: it proves the three detector interfaces stated in the main argument. Appendices B, C, E, and F record $H_3$ refinements that use a chosen $A_4$-structure; Appendix D compares the arithmetic marker data of all three configurations. No canonical identification of the depth plane with the relative-cubic plane is asserted. Appendix G gives the verification architecture for all statements. The optional finite refinements in Appendices B, D, and F are explicitly computer-assisted propositions: their concise tables are article-level outputs, while the complete machine-readable inputs, deterministic algorithms, checksums, and independent replays are the stable supplement listed in Appendix G. They are not presented as unaided human proofs.

# A Detector calculations for uniform exclusion

The first-pass outputs of this appendix are Lemmas 3.2, 3.3, and 3.4. The calculations below prove only those detector channels; no exhaustive description of the finite-group socle is asserted.

Fix $k = \overline{\mathbb{F}}_q$, put $E = L(1)$, and write

$$\mathbf{G} = \mathrm{SL}_2(k), \qquad \Gamma = \mathrm{SL}_2(q), \qquad d = \frac{q-3}{2}, \qquad F = \mathrm{Sym}^d L(2),$$

and set $M = \mathrm{Sym}^2 F$ and $Y_1 = L(1) \otimes L(1)^{(e)}$. Steinberg's tensor-product theorem classifies the simple $\Gamma$-modules used here [12, Theorems 1.1 and 7.4]. Put $D = \nabla(d) = \mathrm{Sym}^d E$. Modular Hermite reciprocity gives the convention-explicit chain

$$\mathrm{Sym}^d(\mathrm{Sym}^2 E) \simeq \mathrm{Sym}^d(\mathrm{Sym}_2 E) \simeq \mathrm{Sym}_2(\mathrm{Sym}^d E) \simeq \mathrm{Sym}^2 D. \tag{A.1a}$$

Here $\mathrm{Sym}_2$ is the lower symmetric square; the first and last identifications are symmetrizations, valid because $2 < p$ [11, Corollary 1.5]. Thus $M \simeq \mathrm{Sym}^2(\mathrm{Sym}^2 D)$; the targeted arguments below do not require a socle formula for the inner symmetric square.

## A.1 The linear detector channels

We first isolate the finite-root seam. Let $S = L(q-1)$ or $L(q-7)$, and let $\phi : S \to F$ be a $\Gamma$-map. For $u(t) : (X, Y) \mapsto (X, Y + tX)$, put

$$D(t) = u_F(t)\phi - \phi u_S(t).$$

The root matrices on $F$ have degree at most $q-3$; those on the two sources have degree at most $q-1$ and $q-7$, respectively. Hence every entry of $D(t)$ has degree at most $q-1$. It vanishes at all $q$ elements of $\mathbb{F}_q$, so it is zero. Thus $\phi$ commutes with the full algebraic positive root group. The finite Weyl element supplies the negative root group, and these two root groups generate $\mathbf{G}$. Consequently, in precisely these two channels,

$$\mathrm{Hom}_\Gamma(S, F) = \mathrm{Hom}_{\mathbf{G}}(S, F). \tag{A.1}$$

This degree argument is deliberately not applied to $M$, whose root degree is $2q-6$.

For $S = \mathrm{St} = L(q-1)$, the top weight of $F$ is $q-3$. Equation (A.1) therefore gives

$$\mathrm{Hom}_H(\mathrm{St}, F) = 0. \tag{A.2}$$

Now suppose $q \equiv 3 \pmod 4$ and $e > 1$. Then $p \equiv 3 \pmod 4$, $e$ is odd, and every base-$p$ digit of $q-7$ is even: from the least significant digit upward they are

$$(p-7, p-1, \ldots, p-1) \quad (p \geq 7), \qquad (2, 0, 2, \ldots, 2) \quad (p = 3).$$

Write $q-7=\sum_j 2r_jp^j$. There is no carry on division by two, so

$$\sum_j r_jp^j = \frac{q-7}{2} = d-2. \tag{A.3}$$

For $2r_j \le p-1$, the submodule generated by $(X^2)^{r_j}$ is the simple module $L(2r_j)=\Delta(2r_j)$, giving a Cartan embedding

$$L(2r_j) \hookrightarrow \operatorname{Sym}^{r_j} L(2).$$

Tensoring its Frobenius twists and multiplying the polynomial factors gives

$$L(q-7) \hookrightarrow \operatorname{Sym}^{d-2} L(2). \tag{A.4}$$

This map is injective: in a monomial basis the three coordinate exponents have base-$p$ digits bounded by $r_j < p$, so different tensor monomials have different exponent triples. Multiplication by the discriminant $Q \in \operatorname{Sym}^2 L(2)$ is injective and completes

$$L(q-7) \hookrightarrow \operatorname{Sym}^{d-2} L(2) \xrightarrow{\cdot Q} F. \tag{A.5}$$

In the covariant $\mathrm{GL}_2$-convention, the source and target degrees in (A.5) are $q-7$ and $q-3$, and $Q$ transforms by $\det^2$. Thus (A.5) is $\mathrm{GL}_2$-equivariant from $L(q-7)\otimes\det^2$ to $F$. After the target is normalized by $\det^{-d}$, the source twist is

$$\det^{2-d} = \det^{-(q-7)/2}. \tag{A.6}$$

Equation (A.1) shows that every finite-group map in this channel is algebraic, so no occurrence can have the other outer normalization. That other extension differs by $\det^{(q-1)/2}$.

At $q=9$, the same construction has a familiar model. Under Hermite reciprocity write $A=X^3, B=X^2Y, C=XY^2, D=Y^3$. The map is the catalecticant-minor covariant

$$X^2 \longmapsto B^2-AC, \qquad XY \longmapsto AD-BC, \qquad Y^2 \longmapsto C^2-BD.$$

It accounts for the $L(2,0)=L(q-7)$ channel; no conclusion about the remaining socle is drawn from this example.

For the prime-field statement, let $q=p\ge 5$. Since $d<p$, the ordinary symmetrizer realizes $F$ as a direct summand of $L(2)^{\otimes d}$. Thus $F$ is tilting. The weight generating function gives the Fischer character identity

$$\operatorname{ch}\operatorname{Sym}^d L(2) = \sum_{0\le j\le\lfloor d/2\rfloor} \chi_{2d-4j}; \qquad \sum_{d\ge 0}\operatorname{ch}(\operatorname{Sym}^d L(2))t^d = \frac{1}{(1-tz^2)(1-t)(1-tz^{-2})}. \tag{A.7}$$

Every highest weight in (A.7) lies between $0$ and $p-3$, where the indecomposable tilting module is simple. Uniqueness of tilting decomposition therefore yields

$$F = \bigoplus_{0\le j\le\lfloor (p-3)/4\rfloor} L(p-3-4j), \tag{A.8}$$

with multiplicity one; every summand has a projection. This proves the targeted linear-detector lemma.

## A.2 The opposite quadratic parity

We now treat maps into $M = \mathrm{Sym}^2 F$; the automatic algebraicity argument above is not available. Let $S = L(s)$ be one of the detector sources and put $X = \mathrm{Hom}_k(S, M)$. The polynomial degree of $M$ is $N = 2q - 6$. With the covariant normalization, a rational map from $S$ uses determinant exponent $(N - s)/2$; after tensoring source and target by their normalizing determinant powers, the two projective extensions differ by $\det^{(q-1)/2}$.

The algebraic-torus weights in a $\Gamma$-fixed element of $X$ are multiples $j(q-1)$, and

$$|\operatorname{wt} X| \le N + s \le 3q - 7 < 3(q-1).$$

Thus a map $\phi$ of the outer parity opposite to the normalized one has a unique decomposition

$$\phi = \phi_+ + \phi_-, \qquad \operatorname{wt}(\phi_+) = q - 1, \quad \operatorname{wt}(\phi_-) = 1 - q. \tag{A.9}$$

For the positive root group define

$$D(t) = u_M(t)\phi - \phi u_S(t).$$

Its entries have degree below $2q$ and vanish on $\mathbb{F}_q$, so

$$D(t) = (t^q - t)B(t), \qquad \deg B < q. \tag{A.10}$$

The cocycle identity gives $B(t + a) = B(t)u_S(a)$, first for $a \in \mathbb{F}_q$ and then for every $a \in k$, since both sides have degree at most $q - 1$ in $a$. Hence, for $\beta_+ = B(0)$,

$$u_M(t)\phi u_S(-t) - \phi = (t^q - t)\beta_+. \tag{A.11}$$

Comparing divided powers gives

$$E^{(1)}\phi_+ = -\beta_+, \qquad E^{(q)}\phi_- = \beta_+, \qquad E^{(1)}\phi_- = E^{(q)}\phi_+ = 0. \tag{A.12}$$

On the weight basis

$$A = x \otimes x^{(e)}, \quad C = y \otimes x^{(e)}, \quad B = x \otimes y^{(e)}, \quad D = y \otimes y^{(e)}$$

of $Y_1$, choose the Weyl element with $wA = D, wC = -B, wB = -C, wD = A$, and define

$$\Theta(A) = -\beta_+, \quad \Theta(C) = \phi_+, \quad \Theta(B) = -\phi_-, \quad \Theta(D) = w\Theta(A). \tag{A.13}$$

The full divided-power table is

$$\begin{array}{c|cccc} & \Theta(A) & \Theta(C) & \Theta(B) & \Theta(D) \\ \hline E^{(1)} & 0 & \Theta(A) & 0 & \Theta(B) \\ E^{(q)} & 0 & 0 & \Theta(A) & \Theta(C) \\ E^{(q+1)} & 0 & 0 & 0 & \Theta(A) \\ \hline F^{(1)} & \Theta(C) & 0 & \Theta(D) & 0 \\ F^{(q)} & \Theta(B) & \Theta(D) & 0 & 0 \\ F^{(q+1)} & \Theta(D) & 0 & 0 & 0. \end{array} \tag{A.14}$$

Indeed, (A.12) gives the two middle columns and Weyl conjugation gives their negative-root counterparts. The ordinary divided-power commutator gives $F^{(1)}\Theta(A) = \Theta(C)$, since $q - 1 = -1$ in $k$. Applying

$$E^{(q)}F^{(q)} = \sum_i F^{(q-i)} \binom{H - 2q + 2i}{i} E^{(q-i)}$$

to $\Theta(C)$ leaves $i = q, q-1$, with coefficients $0, 1$, and gives $F^{(q)}\Theta(A) = \Theta(B)$. Multiplication in the distribution algebra gives $F^{(q+1)}\Theta(A) = \Theta(D)$.

It remains to exclude unlisted divided powers. Use the Chevalley factorization

$$u(s)\ell(t) = \ell\left(\frac{t}{1+st}\right) h(1+st) u\left(\frac{s}{1+st}\right) \tag{A.15}$$

when $1 + st \neq 0$, then clear denominators. Substitution of (A.12) and the three corner coefficients above, followed by comparison of coefficients of $s^a t^b$, gives

$$\ell(t)\Theta(A) = \Theta(A) + t\Theta(C) + t^q\Theta(B) + t^{q+1}\Theta(D).$$

Thus $F^{(i)}\Theta(A) = 0$ for $i \notin \{0, 1, q, q+1\}$; Weyl conjugation and the middle-column relations give every remaining zero in (A.14). The weights, Weyl action, and (A.14) prove that $\Theta : Y_1 \to X$ is a $\mathbf{G}$-map. Moreover $\Theta(C - B) = \phi$. If $\Theta = 0$, then the root defect vanishes and $\phi$ is $\mathbf{G}$-fixed, impossible for the two nonzero weights in (A.9). We have therefore constructed the injection

$$\mathrm{Hom}_{\mathrm{PGL}_2(q)}(S^{\square}, M) \hookrightarrow \mathrm{Hom}_{\mathbf{G}}(S \otimes Y_1, M). \tag{A.16}$$

We verify the right side is zero source by source. For $S = \mathrm{St}$, Doty–Henke's bottom-alcove tensor rule and canonical tilting factorization give

$$\mathrm{St} \otimes L(1) \simeq T(q), \qquad \mathrm{St} \otimes Y_1 \simeq T(2q).$$

These are the bottom-alcove tensor identities of [10, Lemmas 1.3–1.4]. The canonical tilting digits of $2q$ are $(p, p-1, \ldots, p-1, 1)$. The module $T(p)$ has the two Weyl sections $\Delta(p)$ and $\Delta(p-2)$ [10, Lemma 1.1]; hence $T(2q)$ has exactly

$$\Delta(2q), \qquad \Delta(2q-2), \tag{A.17}$$

once each. Both highest weights exceed the top weight $2q-6$ of $M$. Applying $\mathrm{Hom}_{\mathbf{G}}(-, M)$ successively to this two-section Weyl filtration gives

$$\mathrm{Hom}_{\mathbf{G}}(\mathrm{St} \otimes Y_1, M) = 0.$$

For $q = p$ and $S = L(s)$, $s \in \{2, 4, 6, 8, 12\}$ in the ranges of (3.2f), the only simple sources in $S \otimes Y_1$ are $L(s+1, 1)$ and $L(s-1, 1)$. The module $M$ is tilting: $F$ is tilting by (A.8), and the symmetric square is a direct summand of the tensor square because $p$ is odd. Every indecomposable summand $T(n)$ of $M$ has $n \leq 2p-6$, and its restricted simple socle is $L(n)$ for $n < p$, or $L(2p-2-n)$ for $n \geq p$. Neither two-digit candidate embeds, so the Hom space in (A.16) is zero.

Finally take $S = L(q-7)$. If $p \geq 7$, then

$$L(q-7) \otimes L(1) = L(q-6) \oplus L(q-8),$$

with the second term omitted for $p = 7$; tensoring by $L(1)^{(e)}$ gives the sources $L(2q-6)$ and $L(2q-8)$.

We make the plethystic normalization explicit. Write

$$e_i = X^{d-i}Y^i, \qquad a_{ij} = e_i \odot e_j \in \mathrm{Sym}^2 D, \qquad [i, j \mid k, l] = a_{ij} \odot a_{kl} \in M,$$

with the indices within each pair, and then the two pairs, ordered. These are nested symmetric products with no hidden factor of 2; in particular $[0, 0 \mid 3, 3]$ and $[0, 3 \mid 0, 3]$ are distinct. Put

$$X_0 = [0, 0 \mid 0, 0], \qquad Y_0 = [0, 0 \mid 0, 1].$$

For the negative and positive root groups,

$$\begin{aligned} \ell(t)e_i &= \sum_{r=0}^{d-i} \binom{d-i}{r} t^r e_{i+r}, \\ u(t)e_i &= \sum_{r=0}^{i} \binom{i}{r} t^r e_{i-r}; \end{aligned} \tag{A.17a}$$

hence $E^{(1)}Y_0 = X_0$.

For $p \geq 7$, set

$$a = \frac{p-3}{2}, \qquad b = \frac{p-7}{2}.$$

The highest-weight space of $M$ is $kX_0$. A highest vector $g$ of $L(2q-6)$ satisfies $F^{(p-5)}g = 0$, because the least base-$p$ digit $p-5$ of the divided-power index exceeds the least digit $p-6$ of its highest weight. On the target, writing $A(t) = \ell(t)(e_0 \odot e_0)$, the coefficient of $[0,0 \mid b,a]$ in $F^{(p-5)}X_0$ is

$$4\binom{d}{b}\binom{d}{a}. \tag{A.17b}$$

Indeed, $a+b = p-5$; the two factors of 2 are respectively the inner and outer symmetric-square cross terms. Since $d \equiv a \pmod{p}$, Lucas' theorem gives

$$\binom{d}{a} = 1, \qquad \binom{d}{b} = \binom{a}{b} = \binom{a}{2},$$

so (A.17b) is nonzero, including when $p = 7$ and $b = 0$. Thus the image of $g$ must vanish. The weight-$(2q-8)$ space is $kY_0$, but $E^{(1)}Y_0 = X_0 \neq 0$; hence no highest vector of $L(2q-8)$ can map nontrivially either.

If $p = 3$, write

$$L(q-7) \otimes Y_1 = T(3) \otimes R^{(2)},$$

where the previously suppressed higher-digit factor is

$$R^{(2)} = \left( \bigotimes_{j=2}^{e-1} L(2)^{(j)} \right) \otimes L(1)^{(e)};$$

the digit $j = 1$ is trivial. The module $T(3)$ has Weyl sequence $0 \to \Delta(3) \to T(3) \to \Delta(1) \to 0$. Let $g$ be the highest generator of the $\Delta(3)$-section, tensored with the highest vector of $R^{(2)}$, and let $v$ lift the highest generator of the $\Delta(1)$-quotient, normalized by $E^{(1)}v = g$. They have weights $2q-6, 2q-8$ and generate the source. The matching target spaces are spanned by $X_0, Y_0$, so a map has $g \mapsto \alpha X_0,\ v \mapsto \beta Y_0$. But $F^{(6)}g = 0$, while

$$\mathrm{coeff}_{[0,0|3,3]}(F^{(6)}X_0) = 2\binom{d}{3}^2 = 2. \tag{A.17c}$$

Here the factor 2 is the outer cross term; Lucas' theorem gives $\binom{d}{3} = 1$ from the base-3 digits of $d = (3^e - 3)/2$. Hence $\alpha = 0$, and $E^{(1)}v = g,\ E^{(1)}Y_0 = X_0$ give $\beta = 0$. This proves Lemma 3.3.

## A.3 Contraction and the prime-field affine class

Let $i : S \hookrightarrow F$ have an $H$-retraction $\rho : F \to S$. Define

$$D_\rho : \operatorname{Sym}^2 F \longrightarrow S \otimes F, \qquad D_\rho(xy) = \rho(x) \otimes y + \rho(y) \otimes x.$$

Compose $D_\rho$ with a cochain $S \to \operatorname{Sym}^2 F$, then take the categorical trace over $S$. This gives an $H$-map

$$C_{\rho,i} : \operatorname{Hom}(S, \operatorname{Sym}^2 F) \longrightarrow F.$$

For the cocycle $z_i(g)(t) = i(t)c(g)$ from (3.2b), the first summand closes the $S$-loop and the second projects $c(g)$. Therefore

$$C_{\rho,i}(z_i)(g) = (\dim S)c(g) + i\rho(c(g)), \tag{A.18}$$

and the same identity holds in cohomology.

It remains to identify the prime-field class and its support. Put

$$r = \frac{p-1}{2}, \qquad m = r+1, \qquad P_0 = Y(X^r - Z^r) \in R_m, \qquad R_m = k[X,Y,Z]_m.$$

Let $D = s^p t - st^p$ and

$$E_{\text{poly}} = \{P \in R_m : P \circ \nu \in kD\}.$$

With the determinant normalization of (2.2), restriction to the conic is the exact $G$-sequence

$$0 \longrightarrow QR_{m-2} \longrightarrow E_{\text{poly}} \xrightarrow{\epsilon_{\text{poly}}} k \longrightarrow 0, \tag{A.19}$$

where $\epsilon_{\text{poly}}(P) = \lambda$ when $P \circ \nu = \lambda D$. Since the conic ideal is $(Q)$, multiplication by $Q$ identifies the kernel with $F = R_{m-2}$. The affine chart $\epsilon_{\text{poly}} = 1$ is the ambient point-vector extension in (3.2d).

The base form $P_0$ restricts to $D$, and its connecting cocycle is

$$c(u(z)) = \frac{u(z)P_0 - P_0}{Q}, \qquad u(z) : (X,Y,Z) \mapsto (X, Y + zX, Z + 2zY + z^2X).$$

The coefficient of $z^{p-2}$ is

$$[z^{p-2}]c(u(z)) = -rX^{r-1} \neq 0. \tag{A.20}$$

Indeed the numerator coefficient is $-rX^{r-1}Q$: the terms containing one or two copies of $2zY$ sum to $-2r^2X^{r-1}Y^2$, the term containing $Z$ is $-rX^rZ$, and $2r = -1$ in $k$. A coboundary $u(z)A - A$, $A \in R_{m-2}$, has degree at most $p-3$ in $z$, so it cannot cancel (A.20). Thus $[c] \neq 0$.

Finally restrict cohomology to the prime-field Borel. On $L(n)$, the root group $U = C_p$ is one Jordan block, so $H^1(U, L(n))$ is its coinvariant line; the torus acts on that line by $a^{-(n+2)}$. For even $0 \le n \le p-3$, this line is fixed exactly when $n = p-3$. Restriction from $H$ is injective because the Borel index is $p+1$, which is nonzero in characteristic $p$. Hence $[c]$ is supported only on the top Fischer summand in (A.8). Formula (A.18) is therefore $(s+1)[c]$ on a lower detector and $(s+2)[c]$ on the top detector. The ranges in (3.2f) make the first scalar nonzero; in the top case $s = p-3$, so the second is $p-1 \neq 0$. This proves Lemma 3.4.

## B Six profiles and matching-row reconstruction

Specialize to $H_3$. Put

$$G = \mathrm{PGL}_2(11), \qquad G^+ = \mathrm{PSL}_2(11), \qquad X = \Omega_{H_3} \simeq G/H,$$

where $H \simeq A_5$ is the stabilizer of the base matching in (3.1). Thus $X = X_+ \sqcup X_-$, with $|X_+| = |X_-| = 11$. The construction separates the unordered and ordered data:

$$\{A_+, A_-\} \longmapsto K,$$
$$(K, (A_+, A_-),\ \text{convention (B.6)}) \longmapsto \text{oriented cell pairs} \longmapsto D(M) \longmapsto K\backslash G/H.$$

The unordered parent pair supplies the $A_4$-subgroup $K$. The parent ordering together with the displayed orientation convention (B.6) supplies four signed incidence counts; those counts recover the double-coset label. Only the construction of $K$ and the oriented cell pairs requires coordinates.

### B.1 The decorated parent correspondence

On $V = \mathbb{F}_{11}^3$, set

$$q_0(x, y, z) = x^2 + y^2 + z^2, \qquad \mathcal{C}_0 = \mathbb{P}\{q_0 = 0\}.$$

The projectivity

$$\psi = \begin{pmatrix} 10 & 2 & 1 \\ 1 & 2 & 10 \\ 3 & 0 & 3 \end{pmatrix} \quad \text{satisfies} \quad q_0(\psi(X, Y, Z)^t) = 3(XZ - Y^2), \tag{B.1}$$

so $\psi \circ \nu$ identifies the endpoint conic of Section 2 with $\mathcal{C}_0$. In these coordinates take the two six-point sets

$$\begin{aligned} A_+ &= \{(0:1:4), (0:1:7), (1:0:3), (1:0:8), (1:4:0), (1:7:0)\}, \\ A_- &= \{(0:1:3), (0:1:8), (1:0:4), (1:0:7), (1:3:0), (1:8:0)\}. \end{aligned} \tag{B.2}$$

Let $\mathcal{P} = G \cdot A_+$, where $G$ is viewed as the full projective stabilizer of $\mathcal{C}_0$. For $A \in \mathcal{P}$ and $a \in A$, let $C_{A,a}$ be the conic through $A \setminus \{a\}$. The six pairs

$$C_{A,a} \cap \mathcal{C}_0, \qquad a \in A, \tag{B.3}$$

form a perfect matching $M_A$ of the twelve points of $\mathcal{C}_0$. On this 22-point orbit, $A \mapsto M_A$ is a $G$-equivariant bijection onto $X$. This is the *matching-decorated parent correspondence*; its bijectivity and the assertion that (B.3) consists of six disjoint pairs are exact finite inputs. Concretely, the supplement enumerates the 22 translates of $A_+$, deletes each of their six points, solves the resulting five-point conic, intersects it with $\mathcal{C}_0$, rejects unless the six intersections are disjoint two-sets, and compares the resulting canonical matching with all 22 rows of $X$. It is the secant-factorization counterpart of the self-polar matching and Brianchon-support reconstruction obtained from the Clebsch decoding geometry [14].

Put $H_\pm = \mathrm{Stab}_G(A_\pm)$ and

$$K = H_+ \cap H_-, \qquad J = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{pmatrix}. \tag{B.4}$$

Then $K \simeq A_4$, $J(A_+) = A_-$, and $J$ normalizes $K$ and exchanges $X_+$ with $X_-$. Thus the *ordered parent pair* means precisely the ordered pair $(A_+, A_-)$.

Choose linear representatives of $K$ preserving $q_0$, and let $\widetilde{K} = \mathbb{F}_{11}^{\times} K$. For the following eight vectors, define $R_i = \mathbb{P}(\widetilde{K} r_i)$:

$$\begin{array}{cc|cc} i & r_i & i & r_i \\ \hline 1 & (0,1,4) & 10 & (0,1,3) \\ 3 & (0,1,2) & 13 & (0,1,5) \\ 6 & (1,3,2) & 14 & (1,1,5) \\ 9 & (1,4,2) & 11 & (1,2,4). \end{array} \tag{B.5}$$

These are cells in the scalar-$K$ orbit partition of $\mathbb{P}(V)$. The involution $J$ exchanges the four pairs

$$(R_1, R_{10}), \quad (R_3, R_{13}), \quad (R_6, R_{14}), \quad (R_9, R_{11}). \tag{B.6}$$

Only these four oriented pairs enter the argument. By the convention in (B.6), reversing the parent ordering reverses all four orientations.

This profile layer uses more decoration than the balanced-sheet theorem. The quotient configuration intrinsically recovers the unordered sheets, but it does not select an ordered parent pair or the resulting scalar-$K$ refinement. The reconstruction below is relative to that selected refinement.

## B.2 Incidence profiles

For $M \in X$, let $S_M$ be the union of its six secants in $\mathbb{P}^2(\mathbb{F}_{11})$, and put

$$Z_M(R) = |S_M \cap R|.$$

Equivalently, $Z_M(R)$ counts the points of $R$ at which $P_M$ vanishes. Define the *depth profile*

$$\begin{aligned} D(M) = \big(&Z_M(R_1) - Z_M(R_{10}),\ Z_M(R_3) - Z_M(R_{13}), \\ &Z_M(R_6) - Z_M(R_{14}),\ Z_M(R_9) - Z_M(R_{11})\big) \in \mathbb{F}_{11}^4. \end{aligned} \tag{B.7}$$

The entries here and in the profile table below are first formed as integer differences of cardinalities, using the displayed cell-orientation convention in (B.6), and only then reduced modulo 11. This definition uses only zero sets of secant products, so it is unchanged if their equations are rescaled.

**Theorem B.1** (Six-profile reconstruction)**.** *The $K$-orbits on each sheet $X_\pm$ have sizes $1, 4, 6$. With the orientation in (B.6), the corresponding profiles and fibre sizes are*

$$\begin{array}{ccc} \textit{sheet} & |D^{-1}(v)| & v \\ \hline + & 1 & v_1 = (-6, 0, 12, -12), \\ + & 4 & v_2 = (-3, 3, 0, 3), \\ + & 6 & v_3 = (3, -2, -2, 0), \\ - & 1 & -v_1, \\ - & 4 & -v_2, \\ - & 6 & -v_3. \end{array} \tag{B.8}$$

*The six vectors are pairwise distinct. Hence $D$ recovers the double-coset row in*

$$K \backslash G / H$$

*of every matching. Its linear span nevertheless has dimension only two: over $\mathbb{F}_{11}$ it is the plane*

$$2a + 2b + c = 0, \qquad 9a + 8b + d = 0. \tag{B.9}$$

*The two singleton profiles recover their individual matchings. Under the matching-decorated parent correspondence, they therefore recover the corresponding individual $H_3$ parent. A profile in a fibre of size four or six recovers only that $K$-orbit, not an individual matching.*

*Computer-assisted proof mode.* The subgroup-mark reduction and the implications from equivariance are conceptual. The six representative incidence rows in (B.10) below are certificate-checked exact counts and are reconstructed by an independent implementation. A companion certificate verifies that the matching decoration uniquely recovers its parent on the 22-point $H_3$ locus.

*Proof.* The permutation character of $K$ on either eleven-point sheet has values

$$(11, 3, 2)$$

on elements of orders $1, 2, 3$. The marks of the transitive $A_4$-sets are

| $K$-set | size | marks on orders $1,2,3$ |
|---|---|---|
| $K/K$ | 1 | $(1,1,1)$ |
| $K/V_4$ | 3 | $(3,3,0)$ |
| $K/C_3$ | 4 | $(4,0,1)$ |
| $K/C_2$ | 6 | $(6,2,0)$ |
| $K/1$ | 12 | $(12,0,0)$. |

Comparing marks gives the unique nonnegative decomposition

$$X_+|_K \simeq K/K \ \sqcup \ K/C_3 \ \sqcup \ K/C_2.$$

The orbit sizes are therefore $1, 4, 6$; conjugation by $J$ gives the same decomposition on $X_-$.

For $k \in K$, the transformation $k$ preserves every cell in (B.6) and carries $S_M$ to $S_{kM}$. Thus $D(kM) = D(M)$. The involution $J$ exchanges the two cells in every oriented pair and carries $S_M$ to $S_{JM}$, so

$$D(JM) = -D(M).$$

It remains to count one representative on each positive-sheet orbit. The exact incidence rows are

| orbit size | $\mathrm{Stab}_K(M)$ | zero counts in the four pairs of (B.6) | $D(M)$ |
|---|---|---|---|
| 1 | $A_4$ | $(0,6;\ 0,0;\ 12,0;\ 0,12)$ | $v_1$, |
| 4 | $C_3$ | $(3,6;\ 3,0;\ 3,3;\ 6,3)$ | $v_2$, |
| 6 | $C_2$ | $(3,0;\ 1,3;\ 2,4;\ 6,6)$ | $v_3$. |

(B.10)

The $J$-images supply the three negative rows. These six vectors are pairwise distinct, so their fibres are precisely the six $K$-orbits.

Row reduction over $\mathbb{F}_{11}$ gives rank two and the equations (B.9). This rank drop does not merge orbit labels: linear rank and set-theoretic separation are different assertions. Finally, a singleton fibre contains its matching without residual choice. The stabilizer of the matching $M_A$ from (B.3) is the same $A_5$ as the stabilizer of its parent $A$. The equivariant decorated transform between the two $G/H$-orbits is therefore a bijection, so the matching also determines its parent. The stated limitation for the other four rows follows from their fibre sizes. □

**Corollary B.2** (Decorated sheet classifier)**.** *Relative to the orientation in (B.6), the profile $D(M)$ determines the sheet containing $M$. The antipodal singleton pair $\{v_1, -v_1\}$ recovers the unordered matching pair $\{M_{A_+}, M_{A_-}\}$ and hence the unordered parent pair $\{A_+, A_-\}$; choosing the ordering selects one member.*

*Proof.* The positive and negative profile sets in (B.8) are disjoint, and the only singleton fibres are $v_1$ and $-v_1$. Apply the matching–parent bijection from Theorem B.1. □

Although the ambient radial coordinate separates the sheets, $D$ uses additional scalar-$K$ incidence data and is not an affine-linear functional on the abstract configuration. The quadratic and cubic moments remain the intrinsic recovery mechanism of Section 4.

The profile compression also retains the cubic-first orientation from Section 4. The three positive vectors satisfy the integral weighted relation

$$v_1 + 4v_2 + 6v_3 = 0. \tag{B.11}$$

**Corollary B.3** (Orbit weights from normalized profiles)**.** *The three incidence-normalized vectors $v_1, v_2, v_3$ recover the orbit-size multiset $\{1, 4, 6\}$. They also recover the stabilizer orders $\{12, 3, 2\}$ inside $K \simeq A_4$. The underlying rational rays alone do not recover these weights.*

*Proof.* The coordinates of each $v_i$ are fixed by the cardinality differences in (B.7), so independent rescaling is not allowed. These three integral vectors span a plane over $\mathbb{Q}$, and (B.11) is the primitive positive generator of their relation lattice. Its coefficients recover the three orbit sizes without labels. Orbit–stabilizer then gives $12/1, 12/4, 12/6$. If the vectors are replaced by primitive generators of their rays, the relation becomes $1 : 2 : 1$, which proves the final warning. □

# C Modular depth and the projective-cover bridge

Because an $H_3$ sheet has eleven points in characteristic eleven, the sum of its basis vectors is simultaneously constant and augmentation-zero. Linearizing the three incidence rows therefore kills exactly this constant direction and leaves the two-dimensional profile plane. The following module statement identifies that loss.

Retain the notation of Section B, put

$$k = \mathbb{F}_{11}, \qquad L = G^+ = \mathrm{PSL}_2(11), \qquad V = k[X_+] \simeq \mathrm{Ind}_H^L \mathbf{1},$$

and regard $V$ as the permutation module on one eleven-point sheet. Let $e_1, e_4, e_6$ be the indicator functions of the three $K$-orbits, in orbit-size order. Thus

$$V^K = ke_1 \oplus ke_4 \oplus ke_6, \qquad \mathbf{1} = e_1 + e_4 + e_6. \tag{C.1}$$

Linearizing the incidence construction in (B.7) gives

$$\mathcal{D} : V^K \longrightarrow k^4, \qquad e_i \longmapsto i\, u_i, \tag{C.2}$$

where $u_1 = v_1$, $u_4 = v_2$, and $u_6 = v_3$ are the profiles on the orbits of the indicated sizes. Equation (B.11) says exactly that $\mathcal{D}(\mathbf{1}) = 0$.

**Proposition C.1** (Modular depth quotient)**.** *The $L$-module $V$ is the projective cover $P(\mathbf{1})$ and has Loewy dimensions*

$$1 \mid 9 \mid 1.$$

*The map $\mathcal{D}$ has kernel $k\mathbf{1}$ and induces an isomorphism*

$$P(\mathbf{1})^K / \operatorname{soc} P(\mathbf{1}) \xrightarrow{\ \sim\ } \langle u_1, u_4, u_6 \rangle_k. \tag{C.3}$$

*The sheet exchanged by $J$ carries the negative copy of this quotient.*

*Proof mode.* Projectivity, localness, and fixed-point exactness are conceptual. The middle layer follows from the earlier $M_8$-identification. The rank and kernel of $\mathcal{D}$ use the certified profile table (B.8) and its independent replay.

*Proof.* The order of $H \simeq A_5$ is prime to 11, so its trivial module is projective and induction makes $V$ projective. The action of $L$ on $X_+$ is doubly transitive. Hence $\operatorname{End}_L(V)$ is generated by the identity and the all-ones operator $E$. In characteristic eleven,

$$E^2 = 11E = 0,$$

so this endomorphism algebra is local. Thus $V$ is indecomposable. Its trivial head occurs once, and consequently $V = P(\mathbf{1})$. The constant vector is its one-dimensional socle. The unique head map is augmentation, so

$$\operatorname{rad} V = \ker\left(V \xrightarrow{\Sigma} k\right).$$

On this augmentation module define

$$\Phi\left(\sum_{M\in X_+} a_M[M]\right) = \sum_{M\in X_+} a_M\, \pi_8(x_M).$$

Changing the reference matching translates every $x_M$ by the same vector, and the affine cocycle term cancels for the same reason; thus $\Phi$ is a well-defined $G^+$-map. The same-sheet first-moment identity of Proposition 4.1 kills the constant vector and shows that the induced map

$$\overline{\Phi} : \operatorname{rad} V/k\mathbf{1} \longrightarrow M_8$$

is nonzero. The target is the absolutely irreducible $G^+$-module identified in Section 3; both sides have dimension nine. Hence $\overline{\Phi}$ is an isomorphism. This proves the displayed Loewy dimensions without a generator-matrix calculation.

Taking $K$-fixed points is exact because $11 \nmid |K| = 12$. The orbit sizes $1, 4, 6$ give (C.1). The three vectors in (C.2) span the plane (B.9), and their only relation is (B.11), so $\ker \mathcal{D} = k\mathbf{1}$. This proves (C.3). Finally $D(JM) = -D(M)$, which attaches the outer character on the opposite sheet. $\square$

**Corollary C.2** (The canonical nine-space bridge)**.** *For the fixed marked conic and sheet $X_+$, there is a canonical $G^+$-module isomorphism*

$$\operatorname{rad} P(\mathbf{1})/\operatorname{soc} P(\mathbf{1}) \ \xrightarrow{\ \sim\ }\ M_8{\downarrow_{G^+}}\,. \tag{C.4}$$

*It sends the augmentation class of $\sum_{M\in X_+} a_M[M]$ to*

$$\sum_{M\in X_+} a_M\, \pi_8(x_M), \qquad \sum_M a_M = 0, \tag{C.5}$$

*where $\pi_8$ is the top-harmonic projection. Formula (C.5) is independent of the reference matching.*

*Proof.* The construction and all asserted properties of (C.5) are precisely the map $\overline{\Phi}$ established in the proof of Proposition C.1. $\square$

**Corollary C.3** (The homogeneous projective-cover bridge)**.** *For the cocycle representative $c_8$ determined by the chosen reference matching, let $E_{c_8}$ be the $G$-module defined in (3.11b). There is a $G^+$-module isomorphism*

$$P(\mathbf{1})/\operatorname{soc} P(\mathbf{1}) \xrightarrow{\sim} E_{c_8}\!\downarrow_{G^+}, \tag{C.6}$$

*given by*

$$\Theta\left(\sum_{M\in X_+} a_M[M]\right) = \left(\sum_{M\in X_+} a_M\pi_8(x_M),\ \sum_{M\in X_+} a_M\right). \tag{C.7}$$

*Thus the extension class, unlike a cocycle representative or an affine origin, is canonical: the top-harmonic homogeneous extension restricts to the nonsplit exact sequence*

$$0 \longrightarrow M_8\!\downarrow_{G^+} \longrightarrow P(\mathbf{1})/\operatorname{soc} P(\mathbf{1}) \longrightarrow \mathbf{1} \longrightarrow 0.$$

*If $\widetilde{E}_c$ denotes the homogenization of the full affine quotient with translation space $M_8 \oplus M_0$, then*

$$\widetilde{E}_c\!\downarrow_{G^+} \simeq P(\mathbf{1})/\operatorname{soc} P(\mathbf{1}) \oplus \mathbf{1}, \tag{C.8}$$

*where the last summand is the radial module $M_0\!\downarrow_{G^+}$.*

*Proof.* The affine transformation law $\pi_8(x_{gM}) = g\pi_8(x_M) + c_8(g)$ makes $\Theta$ $G^+$-equivariant. The constant vector maps to zero: its second coordinate is $|X_+| = 11 = 0$ in $k$, and its first coordinate vanishes by the same-sheet first-moment identity. Hence $\Theta$ descends through the socle.

On the kernel of augmentation, the descended map is the isomorphism of Corollary C.2; on the one-dimensional head it induces the identity through the second coordinate. It is therefore an isomorphism. The exact sequence cannot split: Proposition C.1 gives $P(\mathbf{1})/\operatorname{soc} P(\mathbf{1})$ the simple socle $M_8$, whereas a splitting would put a trivial summand in its socle. Finally, $M_0 = \chi$ is trivial on $G^+$, and its cocycle vanishes there, proving (C.8). □

*Remark* C.4 (Ambient and intrinsic sheet recovery). With the harmonic–radial decomposition retained, the $M_0$-projection already separates the two $H_3$ sheets: it is zero on $X_+$, constant on $X_-$, and nonzero by Lemma 3.7. The quadratic theorem of Section 4 is stronger in a different sense: it recovers the sheets intrinsically from the abstract affine point configuration, without retaining the ambient Fischer operator.

This quotient identifies the precise modular loss in the six-profile construction. Before linearization, the six profiles remain distinct and recover all six double-coset labels. After linearization, the constant orbit trade dies and only the two-dimensional depth plane remains. Conversely, the three rational rays in that plane remain distinct but do not retain their orbit weights. The weights $1, 4, 6$ are recovered only when the quotient is supplied with the incidence-normalized vectors of (B.8), as in Corollary B.3.

*Remark* C.5 (Reduction of the profile table). The four-coordinate profiles in (B.8) are integral before reduction. The greatest common divisor of their nonzero 2-by-2 minors is 3. Their span therefore drops below rank two only in characteristic 3. The six signed rows remain distinct away from characteristics 2 and 3: characteristic 2 identifies antipodes, while characteristic 3 sends the rows $v_1$ and $v_2$ to zero. This is a statement about the displayed integral table, not a construction of the marked conic geometry in those characteristics.

# D Arithmetic splitting of the marker algebra

The arithmetic representatives of the three configurations come in conjugate pairs. Besides the base matchings in (3.1), we use

$$\begin{aligned} M_{B,-} &= \{\{0,\infty\},\{1,3\},\{2,6\},\{4,5\}\}, \\ M_{B,+} &= \{\{0,\infty\},\{1,5\},\{2,3\},\{4,6\}\}, \\ M_{H,-} &= \{\{0,10\},\{1,\infty\},\{2,7\},\{3,5\},\{4,8\},\{6,9\}\}. \end{aligned} \tag{D.1}$$

The pair for $A_3$ consists of two copies of $M_{A_3}$, the pair for $B_3$ is $(M_{B,-}, M_{B,+})$, and the pair for $H_3$ is $(M_{H_3}, M_{H,-})$. Put

$$f_{A_3} = f_{B_3} = t^2 - 2, \qquad f_{H_3} = t^2 - t - 1;$$

the quadratic marker algebra of type $T$ is $\mathbb{F}_q[t]/(f_T)$. To make the arithmetic bridge part of the definition, let $\bar{k}$ be an algebraic closure of the working field and define the root-specialization maps

$$\begin{array}{ccc} T & \text{root} & \sigma_T(\text{root}) \\ \hline A_3 & \sqrt{2},\ -\sqrt{2} & M_{A_3},\ M_{A_3} \\ B_3 & 3,\ 4 & M_{B,-},\ M_{B,+} \\ H_3 & 4,\ 8 & M_{H_3},\ M_{H,-}. \end{array} \tag{D.2}$$

Thus the *marker datum* is the quadratic algebra together with $\sigma_T$, not the polynomial alone. Exact projective transport places the displayed $B_3$ and $H_3$ values in $\Omega_{B_3}$ and $\Omega_{H_3}$. The certificate also verifies that $\sigma_T$ is the reduction of the corresponding integral Coxeter frames; the theorem below needs only the explicit map (D.2). We call the datum *fused* when its two geometric roots have one rational matching value, and *split* when they give two distinct rational matching sheets.

**Lemma D.1** (Split–inert frame calculation)**.** *The three marker polynomials factor as follows:*

$$\begin{array}{ccc} A_3,\ q=5 & t^2-2 & \textit{irreducible}, \\ B_3,\ q=7 & t^2-2=(t-3)(t-4) & \textit{split}, \\ H_3,\ q=11 & t^2-t-1=(t-4)(t-8) & \textit{split}. \end{array} \tag{D.3}$$

*In the first row the two geometric roots form one Frobenius orbit over $\mathbb{F}_{25}$. In each split row the nontrivial involution of the quadratic marker algebra exchanges the two rational roots.*

*Proof.* The nonzero squares modulo 5 are 1 and 4, so 2 is not a square. If $u^2 = 2$ in $\mathbb{F}_{25}$, the nontrivial $\mathbb{F}_5$-automorphism sends $u$ to $-u$, and hence exchanges the two roots. The remaining factorizations follow by direct substitution. □

**Theorem D.2** (Marker specialization and $H_3$ arithmetic gluing)**.** *For these arithmetic representatives of the rank-three matching configurations, the following hold.*

(a) *The $A_3$ marker datum is fused over $\mathbb{F}_5$. The $B_3$ and $H_3$ marker data split into two rational matching sheets over $\mathbb{F}_7$ and $\mathbb{F}_{11}$, respectively.*

(b) *In the $H_3$ case, the stabilizers $H_+$ and $H_-$ of the singleton matching in each rational sheet are isomorphic to $A_5$, satisfy*

$$H_+ \cap H_- = K \simeq A_4, \qquad \langle H_+, H_- \rangle = \mathrm{PSL}_2(11), \tag{D.4}$$

*and are exchanged by a projectivity $J$ of nonsquare determinant. The normalizer $N_G(K)$ is a rational $S_4$ whose determinant-square kernel is $K$.*

(c) *The determinant character that distinguishes the two rational sheets also negates the depth profiles. Thus choosing one root in the split marker algebra chooses one sheet, while its singleton profile chooses the corresponding decorated matching row. Forgetting that choice leaves the unordered pair.*

*Computer-assisted proof mode.* Lemma D.1 and the determinant-character deductions are conceptual. The agreement of the specialization map (D.2) with the integral Coxeter frames, the matching stabilizers and intersections, and the generated subgroup in (D.4) are exhaustive exact finite checks. The classical subgroup names use the standard subgroup structure of $\mathrm{PGL}_2(11)$ [1, 7].

*Proof.* Part (a) records the specialization data in (D.2). Its two $A_3$ roots have the same matching value. The two values in the $B_3$ row are exchanged by $x \mapsto -x$, and the two values in the $H_3$ row are exchanged by

$$x \longmapsto \frac{x-1}{x+1}.$$

The exact transport check verifies these identities on the displayed matchings. Together with Lemma D.1, they prove (a).

For $H_3$, exact projective closure gives

$$|H_+| = |H_-| = 60, \qquad |H_+ \cap H_-| = 12, \qquad |\langle H_+, H_- \rangle| = 660.$$

Both stabilizers lie in the determinant-square subgroup, while the displayed transporter has nonsquare determinant. The normalizer $N_G(K)$ has order 24, and its determinant-square kernel has order 12. The cited subgroup identifications give (D.4) and the $S_4/A_4$ hinge, proving (b).

The determinant-square subgroup has the two matching sheets as its orbits, and $J$ exchanges them. Section B proves $D(JM) = -D(M)$ and identifies the unique singleton in each sheet. Theorem B.1 then returns the decorated matching row from that singleton. This proves (c). ☐

Arithmetic splitting and modular depth retain different amounts of information. Splitting produces a two-point choice of sheet. The depth quotient is defined only after the $A_4$-refinement has been selected and remembers all three orbit rays modulo the constant socle. The first choice orients the second, but does not canonically identify it with any cubic quotient of the ten-dimensional factorization space. Appendix F makes that boundary precise.

# E Cubic structure of the profile plane

**Lemma E.1** (Three-ray cubic forcing)**.** *Let $k$ have characteristic different from* 2 *and* 3. *Suppose $u_1, u_2$ form a basis of a two-dimensional $k$-space and $n_1, n_2, n_3 \in k^\times$ satisfy*

$$n_1 u_1 + n_2 u_2 + n_3 u_3 = 0.$$

*For the signed antipodal measure*

$$\xi = \sum_{i=1}^{3} n_i (\delta_{u_i} - \delta_{-u_i}),$$

*the moments of degrees one and two vanish, while the cubic moment is nonzero.*

*Proof.* The relation kills the first moment, and antipodality kills every even moment. Write

$$u_3 = -\frac{n_1}{n_3}u_1 - \frac{n_2}{n_3}u_2.$$

In $\sum_i n_i u_i^{\odot 3}$, the coefficient of $u_1^{\odot 2} \odot u_2$ is

$$-\frac{3n_1^2 n_2}{n_3^2},$$

which is nonzero. The signed cubic is twice this tensor and is therefore nonzero as well. $\square$

**Corollary E.2** (The mass-zero cubic flag)**.** *In Lemma E.1, suppose also that $n_1+n_2+n_3 = 0$. Then its half-cubic factors as*

$$\frac{1}{2}M_3(\xi) = \sum_{i=1}^{3} n_i u_i^{\odot 3} = \frac{n_1 n_2}{(n_1+n_2)^2}(u_1-u_2)^{\odot 2} \odot ((2n_1+n_2)u_1 + (n_1+2n_2)u_2). \tag{E.1}$$

*The two displayed lines are distinct. Hence the cubic intrinsically defines a doubled line and a residual simple line; its Hessian recovers the doubled line.*

*Proof.* Since $n_3 = -(n_1+n_2)$, the weighted vector relation gives

$$u_3 = \frac{n_1 u_1 + n_2 u_2}{n_1 + n_2}.$$

Substitution and expansion give (E.1). The residual factor would equal the doubled factor only if $3(n_1+n_2) = 0$, which is impossible under the hypotheses. The Hessian of a binary cubic $L^2R$, with $L$ and $R$ distinct, is a nonzero scalar multiple of $L^2$. $\square$

For the incidence-normalized profiles of Section B, put

$$\eta = \delta_{v_1} + 4\delta_{v_2} + 6\delta_{v_3} - \delta_{-v_1} - 4\delta_{-v_2} - 6\delta_{-v_3}.$$

Then

$$M_j(\eta) = (1-(-1)^j)\sum_{i=1}^{3} n_i v_i^{\odot j}, \qquad (n_1,n_2,n_3) = (1,4,6). \tag{E.2}$$

Equation (B.11) kills the first moment, and antipodality kills every even moment. The first two profiles are independent over $\mathbb{F}_{11}$, since the determinant of their first two coordinates is $-18 \neq 0$. Lemma E.1 therefore forces the cubic to be nonzero. In the frozen normalization, its first coordinate is

$$2((-6)^3 + 4(-3)^3 + 6(3)^3) = 6 \neq 0 \quad \text{in } \mathbb{F}_{11}. \tag{E.3}$$

Thus the six-row quotient preserves the cubic-first signed moment even though its linear image is only a plane.

*Remark* E.3 (The $H_3$ cubic flag)*.* Here $1+4+6 = 0$ in $\mathbb{F}_{11}$, so Corollary E.2 forces the double-line factorization. In the basis $e_1 = v_2, e_2 = v_3$ of the profile plane, the projective cubic is

$$f(x,y) = x^3 + 7x^2y + 5xy^2 + 9y^3 = (x-y)^2(x-2y),$$

with Hessian $7x^2 + 8xy + 7y^2 = 7(x-y)^2$. Hence the cubic intrinsically distinguishes a doubled line and a residual simple line. This flag is internal to the decorated profile plane: it supplies no canonical map from the full relative-cubic space and no descended $G^+$-action.

# F The relative-cubic Tate plane

Let $W = W_{H_3}$ be the ten-dimensional quotient space from Theorem 3.9, let $\chi$ be the determinant character of $G = \mathrm{PGL}_2(11)$, and put

$$M = \mathrm{Sym}^3(W) \otimes \chi, \qquad R = M^G.$$

The signed cubic of Section 4 lies in $R$, which has dimension three. There is a canonical two-dimensional quotient of this source space, but it is not canonically the depth plane of Proposition C.1.

We fix the monomial basis of $\mathrm{Sym}^3(W)$ induced by the harmonic–radial coordinates of Section 3. The *frozen relative basis* is the reduced-echelon basis of the simultaneous equations $gf = \chi(g)f$ for the standard generators defining the marked $H_3$ orbit, ordered by its pivots. Thus the coordinates below specify a reproducible basis, not an additional intrinsic structure. The *three labelled cubic orbit classes* are indexed, in order, by the three $K$-orbits of sizes $1, 4, 6$ on $X_+$. Finally, the *invariant symmetric-form pencil* is $\mathbb{P}((\mathrm{Sym}^2 W)^G)$. Its unique rank-one point has the form $[\ell^2]$; contraction by its covector $\ell$ is the induced map

$$\iota_\ell : R \longrightarrow (\mathrm{Sym}^2 W)^G.$$

**Proposition F.1** (Canonical relative-cubic Tate plane)**.** *The invariant space $R$ and the coinvariant space $M_G$ both have dimension three. The canonical projection and group norm fit into*

$$R \xrightarrow{\pi} M_G \xrightarrow{N} R, \qquad \mathrm{rank}\,\pi = 2, \quad \mathrm{rank}\,N = 1, \tag{F.1}$$

*with*

$$\mathrm{im}\,\pi = \ker N, \qquad \mathrm{im}\,N = \ker \pi = \langle [1:3:9] \rangle. \tag{F.2}$$

*Contraction by the covector whose square is the unique rank-one member of the invariant symmetric-form pencil gives a second construction of the same quotient. In the frozen relative basis its matrix is*

$$\begin{pmatrix} 8 & 1 & 0 \\ 0 & 8 & 1 \end{pmatrix}, \tag{F.3}$$

*and its kernel is again $\langle [1:3:9] \rangle$.*

*Computer-assisted proof mode.* The invariant/coinvariant construction and the implications of the displayed ranks are conceptual. The dimensions, norm and projection ranks, kernel line, rank-one pencil member, contraction matrix, and flattening census are exact finite linear algebra in the `verification/evidence/relative_cubic_depth.json` bundle, generated by `relative_cubic_depth.py` and checked by its independent replay. The fingerprint in Appendix G fixes all three files and their exact inputs.

*Proof.* Exact invariant and commutator calculations give $\dim R = \dim M_G = 3$, the two ranks in (F.1), and $N\pi = 0$. Thus $\mathrm{im}\,\pi \subseteq \ker N$; both spaces have dimension two, so they are equal. The computed image of $N$ is the displayed kernel of $\pi$, proving (F.2).

The invariant symmetric-form pencil on $W$ has a unique rank-one member $\ell^2$, where $\ell$ transforms by $\chi$. Contracting a $\chi$-twisted cubic by $\ell$ cancels the two characters and lands in the invariant quadratic pencil. Exact contraction gives (F.3), whose kernel proves the second assertion. □

The kernel line is intrinsic rather than an artifact of the displayed basis. Among the 133 projective lines of $R$, it is the unique line whose first flattening has rank 9; the remaining census is one line of rank 1 and 131 lines of rank 10.

*Remark* F.2 (Why the two canonical planes do not glue). The three labelled cubic orbit classes project to $M_G$ with relation $[2,9,1]$, whereas the three labelled depth profiles have relation $[2,8,1]$. There is therefore no label-preserving linear map carrying these source classes to the depth rays. The integral transfer makes the obstruction structural. In orbit-size order set

$$s = (1,1,1)^t, \qquad B = s(1,4,6).$$

Then $B^2 = 11B$. Divided transfer kills every integral balanced relation $a$ with $(1,4,6)a = 0$, but $Bs = 11s$, so it fixes the depth socle after division by 11. It separates the source relation from the depth socle instead of identifying them. The doubled-line and residual-line flag of Remark E.3 is internal to the depth plane and supplies no missing map.

# G Verification architecture

Table 1 separates the mathematical arguments from their exact finite inputs. Here "certificate" means a deterministic calculation over a prime field, checked against a canonical JSON output and a SHA-256 manifest; an independent replay uses a separate implementation. "Lean" means that the stated implication is checked by the Lean kernel. Neither term replaces the cited identification of the classical groups or configurations.

| Result | Proof in the text or classical input | Exact evidence |
|---|---|---|
| Matching-secant quotient | unique conic divisibility and the four-endpoint switch | none |
| Balanced-orbit completeness | projective–trade bridge, Faber tame subgroups, targeted linear detectors, root-defect injection, and Steinberg/Fischer tests | structural Lean corroboration and a $q = 9$ replay; neither replaces the all-field detector proofs in Appendix A |
| Ranks and the middle Fischer layer | affine cocycle, irreducible Fischer modules; Edge–Dye configurations | quotient matrices as cross-checks; all edge-selected alternating cycles and independent Dickson replay |
| Balanced sheets and cubic orientation | Propositions 4.1 and 4.2; Lemma 4.5; maximal-isotropic quotient duality | finite moments and ranks as cross-checks; kernel-checked hyperplane-square implication; Gorenstein replay |
| Fixed-line ambiguity and Chow rigidity | invariant-space dimensions, subgroup normalizers, unique block systems, and radial translation | structural Lean gate for the radial-translation implications; the coalescence-point square rank is outside the theorem |
| Six-profile reconstruction | subgroup marks, orbit–stabilizer, and the three-ray lemmas | profile rows and replay; decorated-parent bijection |
| Modular depth and the Tate plane | projectivity, fixed-point exactness, and standard modular-module input | invariant, coinvariant, and depth calculations with replay |
| Arithmetic splitting and gluing | displayed root calculations; cited subgroup identifications | Lean gate and normalized certificate/replay |

Table 1: Proof modes for the principal results.

The claim-by-claim map is `verification/trust_manifest.json`; the exact statement identity is `verification/statement_identity.json`. From the root of the archival source, the aggregate entry point is

```
python3 verification/verify_release.py
```

It checks the statement identity and all admitted checksum manifests, runs the admitted primary checks and independent replays, elaborates the structural, arithmetic-gluing, Hilbert-symmetry, and hyperplane-square Lean gates, and rebuilds this manuscript with the pinned manuscript Nix environment and build checker.

The reproducible review source is pinned by

`verification/evidence_fingerprint.json`.

Its SHA-256 digest is

`c2732e7c102c9be489590c76c6f1c37409ff58`
`8ab42445a1890244ec1d433901`.

That fingerprint fixes a normalized manuscript, statement contents, extractor, paper Nix inputs, manuscript checker, adversarial boundary checker, verification documentation, the admitted evidence manifests, command vectors, Lean gates and their full project-owned import closure, runner, trust manifest, and environment: Python 3.13.12, Lean 4.32.0-rc1, and Mathlib revision

`571b8a8e54219b4d393f75f4b8653fac08197fcc`.

A successful replay ends with

`clebsch factorization release:  CHECK OK`.

The normalization replaces only the displayed fingerprint digest and the statement identity's derived full-source hash, avoiding a circular self-hash. This is an explicit review-source allowlist, not a claim to hash every file in the repository or every transitive tool dependency. The fingerprint identifies the source closure; it is not an immutable public locator.

The structural gate

`RelativeConicArcs.Gates.`
`ClebschPaperIIStructural`

follows the proof spine from the projective pullback through the concrete finite-root and outer-defect abstractions, the divided-power detectors, and the affine contraction, to the rank-three endpoint. The all-field representation and subgroup assertions in Appendix A remain human proofs. Its twenty-nine audited terminals include the derived fused $q = 5$ exclusion and point-stabilizer uniqueness certificates for the $B_3/\mathbb{F}_7$ and $H_3/\mathbb{F}_{11}$ matchings. The $H_3$ uniqueness check uses twelve explicit point-stabilizer witnesses per sheet, not a matching census. The same gate imports

`ClebschFixedLineRadialTranslation`.

The exact terminals are

`annihilates_hadamardSquare_iff_`
`eq_sheetSignLine_of_noncoalescent` and `card_nonmatchingNoncoalescentParameters`.

They check the table-free Radical–Hadamard inheritance and the exact $q-2$ count after removing the matching and coalescence parameters. The inheritance is derived, not assumed: the evaluation space is presented as the parameter-independent top space together with the constant function and the radial level, the two radial levels at parameters with nonzero outer constant span the same line, and the three Radical–Hadamard hypotheses are therefore imposed only at the reference parameter. The gate terminal `evaluationSpace_eq_reference` records that step. Identifying the geometric evaluation spaces with this presentation remains a human input. The invariant-space, stabilizer, block-system, and unique-Chow-point arguments remain human and classical inputs. Steinberg, Doty–Henke, and modular Hermite reciprocity supply the cited representation-theoretic interfaces; the printed tilting, root-defect, and contraction arguments prove the detector consequences. Faber supplies the abstract tame subgroup types, while Dickson–Giudici enters only in the later exact-order maximal-overgroup step. These are classical inputs rather than consequences of the Lean gate.

The largest formal component is the gate

```
RelativeConicArcs.Gates.
ClebschArithmeticGluing.
```

Its twenty-three terminals check the displayed roots, matching involutions, projective orders, determinant halves, the $A_3/B_3$ orbit calculations, and the split–fused conclusion. For $H_3$, Lean checks the normalized certificate rows, intersections, determinant classes, the $11 + 11$ signature split, and the shape of the bounded generation words. Evaluation and completeness of the 660-element coset and word certificates remain exact generator/replay obligations. The classical names $S_4, A_4, A_5$, and $\mathrm{PSL}_2(11)$ are cited inputs: no group name is inferred from an order.

The separate structural gate

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RelativeConicArcs.Gates.
ClebschHilbertSymmetry
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retains an independent check of the Hilbert arithmetic: a symmetric finite Hilbert function of total length $2q$ whose first values are $1, q-1, q-1$ has socle degree three and final value one. The Gorenstein proof does not use this implication; it obtains the perfect degree-1-by-degree-2 pairing directly from maximal isotropy.

The other computations are load-bearing only at the points shown in Table 1. The matching-module bundle supplies the three finite quotient ranks and cross-checks the moment data. The $H_3$-equivariant-rank bundle checks the Sylow-normalizer weights, $A_5$-fixed dimensions, sheet membership, and the top witness. The small-field trade bundle exhausts $q = 9$; its larger-field rows are cross-checks only. The shared-radial bundle reconstructs every edge-selected pair, its alternating cycle, and its deepest trace; an independent replay starts from the Dickson recurrence. The balanced-sheet bundle independently reconstructs the permutation-module conclusions, while the Gorenstein bundle checks the ideals, deletion ranks, socles, and resolutions. The profile-incidence bundle computes one row per double coset; the decorated-parent bundle identifies a singleton matching with its parent; the relative-cubic-depth bundle computes the modular and Tate rank patterns; and the arithmetic-gluing gate checks the normalized splitting data. All use exact prime-field arithmetic. The balanced-orbit classification uses no field census: its executable component checks the local formulas after the human modular identifications. The general divisibility theorem, radical–Hadamard and hyperplane-square lemmas, modular-permutation mechanism, three-ray cubic lemma, and mass-zero factorization are proved in the text.

This verification surface proves only the three marked configurations defined here. It does not supply an all-characteristic construction or a canonical map from the relative-cubic Tate plane to the depth plane.

## AI assistance disclosure

This project was developed with extensive assistance from OpenAI Codex and Anthropic Claude. Under the author's direction, the systems assisted throughout with proof exploration, literature research, symbolic and finite computation, code and formal-proof development, verification, and manuscript drafting and revision. The author checked the resulting arguments, computations, code, and cited sources, reviewed and edited all AI-assisted material, and assumes responsibility for all content.

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